Ocean Engineering 270 (2023) 113681

Contents lists available at ScienceDirect

Ocean Engineering

journal homepage: www.elsevier.com/locate/oceaneng

Nonlinear URANS model for path-following control problem towards autonomous marine navigation under wave conditions

Daejeong Kim [a], Soonseok Song [b,*], Stephen Turnock [c], Tahsin Tezdogan [c]

[a] Department of Naval Architecture, Ocean and Marine Engineering, University of Strathclyde, 100 Montrose Street, Glasgow, G4 0LZ, UK
[b] Department of Naval Architecture & Ocean Engineering, College of Engineering, Inha University, 100 Inha-ro, Incheon, Republic of Korea
[c] Department of Civil, Maritime and Environmental Engineering, School of Engineering, University of Southampton, Boldrewood Innovation Campus, Building 176, SO16 7QF, UK

ARTICLE INFO

Handling Editor: Prof. A.I. Incecik

Keywords:
Path-following control
MASS
Maritime autonomous surface ships
LOS
Line-of-Sight
CFD
Free-running ship

ABSTRACT

An accurate prediction of the performance of maritime autonomous surface ships (MASS) to follow a predefined path in waves is of critical importance for ensuring safe autonomous marine navigation. Traditional methods for the study of path-following problems are based on simplified mathematical ship models and are incapable of precisely resolving the complicated interactions between the hull, propeller, rudder, and incident waves under path-following control. In the present study, a CFD-based dynamic model for path-following problems is developed for autonomous marine navigation by means of a fully nonlinear unsteady Reynolds-Averaged Navier-Stokes (RANS) solver combined with the Line-of-Sight (LOS) guidance law. Fortunately, an unsteady RANS solver is capable of incorporating viscous and turbulent effects and the free surface resolution critical to path-following problems, allowing a better prediction of a ship's path-following performance in waves. To obtain practical insight into the performance of the ship executing path-following control for autonomous navigation in waves, a comprehensive analysis of path-following tasks in waves covering the whole range of important wave directions and wave heights was carried out in this study. The numerical results demonstrated that when the ship performed the path-following manoeuvre under the beam and quartering seas, the greatest oscillatory deviation from the predefined path was observed due to the asymmetric pressure distribution on the ship hull caused by the incident waves. It was also confirmed that wave heights strongly affected the trajectories experienced by the ship performing the path-following task in the beam and oblique seas; the deviation from the predefined path was found to increase with the increase in the wave height. As high-performance computational resources become increasingly available, the proposed CFD method will provide an accurate and efficient way to predict the performance of autonomous surface ships performing path-following tasks in waves, providing a valuable contribution to enhancing the safety of autonomous marine navigation.

1. Introduction

Before a voyage begins, the route of a ship is generally determined, taking into account both the important safety factors associated with the route features and a ship's manoeuvring performance. In other words, the route a ship will navigate on is designed to avoid the obstacles or limited water depth areas that can pose a threat to the safety of navigation, as much as possible. The capability of a vessel to follow a predefined path in a real seaway is therefore critical for ensuring safe navigation, which has also been regarded as a prerequisite for the development and deployment of remote-controlled ships or maritime autonomous surface ships (MASS). It has been observed in real operations that a vessel can be subjected to external disturbances such as waves, wind, and currents. Waves, in particular, are a major cause of deviations from the planned route due to the effect of wave loads on a ship's manoeuvring behaviour. Such deviations may consequently lead to an increased sailing distance with financial ramifications and serious maritime incidents including grounding and collision. Therefore, it is imperative to fully evaluate and understand the behaviour of a vessel performing path-following manoeuvres in real sea states in order to make proper voyage planning decisions and ensure safe autonomous navigation operations of ships. The main aim of this study is to develop a Computational Fluid Dynamics (CFD) based model capable of accurately assessing the performance of an autonomous ship performing path-

* Corresponding author.
E-mail address: s.song@inha.ac.kr (S. Song).

https://doi.org/10.1016/j.oceaneng.2023.113681
Received 23 October 2022; Received in revised form 21 December 2022; Accepted 9 January 2023
Available online 19 January 2023
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following manoeuvres in various wave conditions. It is believed that the model developed in this work can also provide a practical insight into understanding the path-following capability of a ship for autonomous navigation in a real seaway.

The Line-of-Sight (LOS) guidance algorithm has been widely used in path-following applications of surface vessels due to its properties in generating appropriate heading reference trajectories. For instance, Fossen et al. (2003) implemented a geometric assignment based on the LOS guidance law for the straight line path-following manoeuvre of a marine craft, demonstrating its capability in minimisation of the cross-track error to the path. The investigation in Borhaug and Pettersen (2005) was focused on a cascaded systems approach in which the LOS guidance algorithm is combined with a sliding model controller to guarantee global κ -exponential stability of the cross-track error to a straight line in 3-D space. In Lekkas and Fossen (2012), a path-following analysis was carried out by means of a modified LOS guidance algorithm using a time-varying lookahead distance (depending on the cross-track error), which can help diminish the oscillatory behaviour around a predefined path. Numerical simulations of path-following in regular waves were performed in Sandeepkumar et al. (2022), where a 3 Degree of Freedom (3DOF) unified manoeuvring model of the KVLCC2 and the LOS algorithm were adopted. Like the aforementioned papers, the present study utilised the LOS guidance algorithm.

A ship manoeuvring model, which has the capability of accurately predicting the manoeuvring behaviour of a ship in response to the rudder deflection in a free surface environment, is also necessary to tackle the path-following problem numerically. It has to be stated that numerous past studies in the field of path-following have been devoted to using a simplified mathematical ship model (such as Manoeuvring Modelling Group (MMG) model (Inoue et al., 1981; Yasukawa and Yoshimura, 2015)) to solve the path-following problem (Fossen et al., 2003; Borhaug and Pettersen, 2005; Lekkas and Fossen, 2012; Sandeepkumar et al., 2022; Zhang et al., 2022). The main disadvantage of the simplified mathematical model lies in the fact that the accuracy of the ship model is strongly reliant on the validity of the hydrodynamic coefficients that need to be acquired from experiments or CFD, which may limit the applicability of this approach (Mofidi and Carrica, 2014). Furthermore, the simplified mathematical model is incapable of precisely resolving the complicated interactions between the hull, propeller, and rudder during manoeuvring (essential to estimate a ship's manoeuvring behaviour) in a given environment. Given this, it can be claimed that the simplified mathematical model is not appropriate for accurately estimating the performance of an autonomous ship performing path-following manoeuvres in a real seaway.

In the Final Report and Recommendations of the 29th ITTC of the Manoeuvring Committee (ITTC, 2021), a free-running CFD simulation has been recommended as the state-of-the-art method for accurately predicting the manoeuvring performance of a ship, due to the capability of incorporating viscous and turbulent effects and the free surface resolution which are critical to the manoeuvring problems. With advances in high-performance computing systems and numerical algorithms, free-running CFD simulations are gaining more and more popularity for the prediction of a ship's manoeuvring behaviour by using unsteady Reynolds-Averaged Navier-Stokes (URANS) computations coupled with the equations of six degree-of-freedom rigid body motions. However, due to its brevity, the application of the free-running CFD model has been only limited to the evaluation of turning and zigzag manoeuvres (Mofidi and Carrica, 2014; Broglia et al., 2015; Dubbioso et al., 2016; Wang and Wan, 2018; Hasanvand and Hajivand, 2019; Hasanvand et al., 2021; Kim et al., 2021a; Kim et al., 2021b; Kim et al., 2021c; Kim et al., 2022a; Kim and Tezdogan, 2022; Kim et al., 2022b, c; Kinaci and Ozturk, 2022). Given this background, the present study was motivated to develop a path-following model under regular wave conditions for autonomous navigation systems, using a fully nonlinear URANS model allowing a better prediction of a ship's manoeuvring performance in response to the rudder behaviour. To the best of our knowledge, there has been no previous published studies concerning a free-running CFD model for path-following in real sea states.

In the present paper, a CFD-based model of path-following in waves was developed for the well-known KCS container ship by a combination of a fully nonlinear unsteady RANS solver and the LOS guidance algorithm. In addition, a complete analysis was provided for the path-following manoeuvres in waves covering a range of important wave directions and heights, to gain more practical insight into understanding of the path-following performance of the ship for an autonomous navigation system in waves. It will be shown that the results confirm and demonstrate the contribution of waves to the ship trajectory, critical parameters related to the LOS guidance, and kinematic parameters experienced by the ship during the path-following manoeuvres.

The remaining paper is framed as follows: Section 2 provides the research methodology of this study, with details presented in the subsections that follow. Section 3 illustrates in detail the CFD results obtained from the free-running simulations for path-following. Finally, Section 4 discusses the findings drawn from this work, as well as recommendations for future research.

2. Methodology

This section will outline the general methodology used in this study, with a detailed explanation of the procedure provided in the contained sub-sections. As seen in Fig. 1, the research methodology applied in this work was organised into three key steps: 1) *goal and scope*, 2) *modelling*, and 3) *results of analysis*. The first step is proposed to determine the overall research goal and the scope of the analysis. The scope of the numerical simulations should be sufficiently well defined to represent the general observations on the relation between the selected factor and the path-following performance of a ship. The scope includes the following items.

- A CFD solver for the free-running simulation.
- Ship types and geometries to be studied.
- Approach conditions including the target speed of a ship
- Environmental conditions such as deep unrestricted water, regular waves, irregular waves, shallow water, and ocean currents.
- Assumptions to be applied.

The numerical modelling of a free-running CFD model for path-following is performed in the second step. Special care is needed when carrying out the numerical modelling task, and in this regard, the following features should be taken into consideration.

- Trajectory generation
- Guidance system algorithm to be adopted
- Control system design
- Free-running CFD model free to move in 6 DOF

After the development of the CFD model for path-following, the self-propulsion computation in a given environmental condition should be first performed to reach the target approach speed. Then, path-following manoeuvres aiming at following the pre-defined route are performed from the stable state of the self-propulsion condition. The third step illustrates and discusses in detail all of the results from this study, in terms of path-following performances, seakeeping characteristics, and manoeuvring behaviours.

2.1. Step 1: Goal and scope

The primary goal of this study is to develop a CFD-based model of path-following for the KCS using a URANS approach and to enhance a comprehensive understanding of the path-following performance for autonomous navigation systems in regular waves.

As depicted in Fig. 2, a 1/75.24 scale model of the KCS characterised

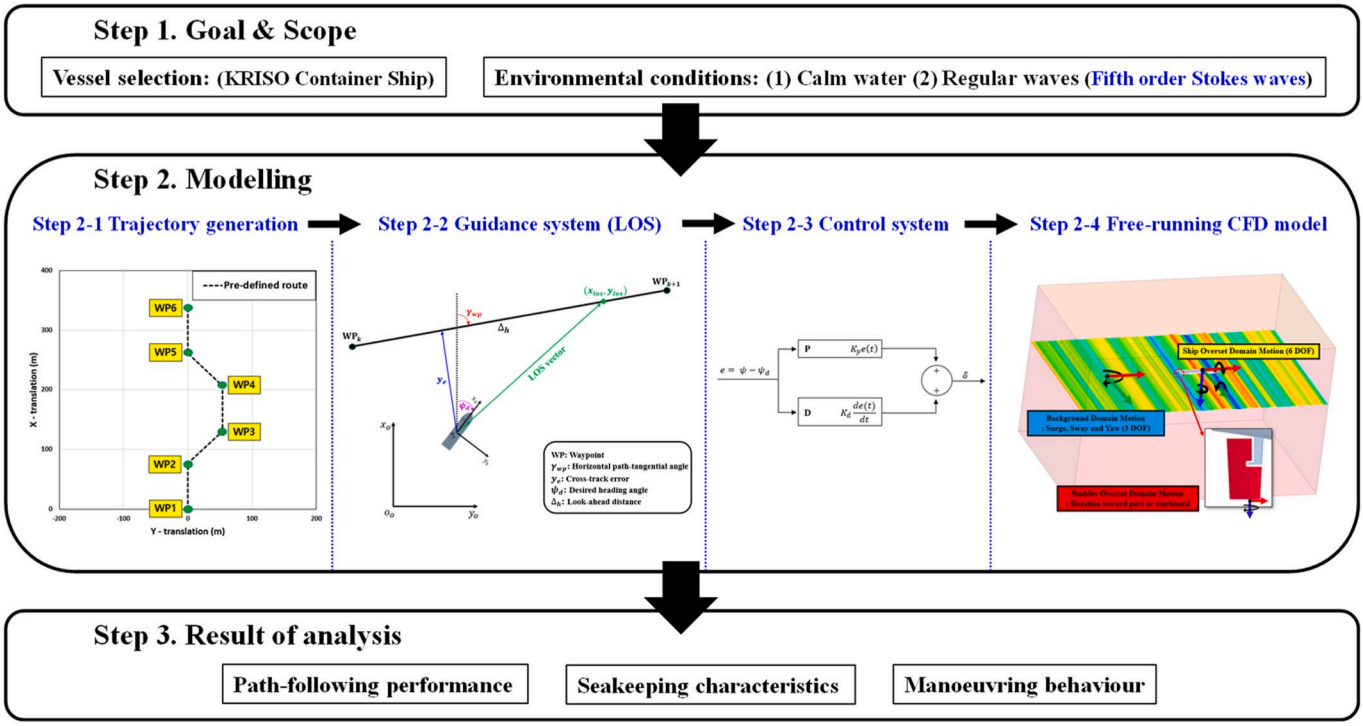


Fig. 1. Research methodology applied to the CFD free-running simulation for path-following.

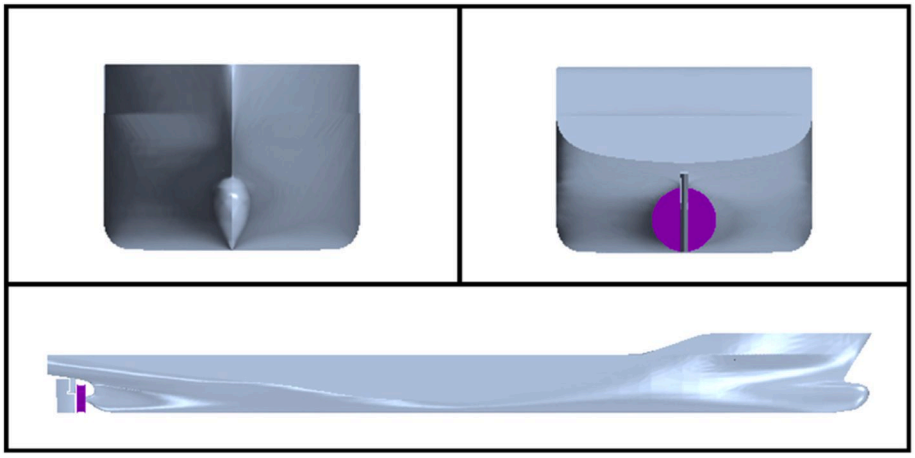


Fig. 2. KCS geometry with a semi balanced skeg rudder and an actuator disk.

by a traditional single rudder/single propeller configuration has been used for the numerical simulation in this work. The principal properties of the KCS model are listed in Table 1. The current study included eleven different simulation cases, as detailed in Table 2, for which the free-running CFD model for path-following was used. For the sake of clearness, the schematic view of the simulation cases is also displayed in Fig. 3. Case 1 was designed to identify the inherent performance of the ship performing the path-following manoeuvre in deep unrestricted water. Cases 2–11 were intended to assess the contribution of waves to the path-following performance of the ship, covering a range of important wave directions and wave heights. Throughout all simulations in this work, the propeller revolution speed was set to be 13.38 RPS, following the experimental setup provided in Yasukawa et al. (2021). It has to be mentioned that the authors' previous work (Kim et al., 2021c) extensively validated the free-running CFD model adopted in this work in terms of kinematic and important manoeuvring parameters using the propeller revolution of 13.38 RPS. When it comes to Cases 2–6, the effects of wave directions (i.e., head, bow, beam, quartering, and following waves) on the path-following manoeuvre were investigated (a wave with height $H = 0.048\ m$ and period $T = 1.4\ s$ in model-scale, corresponding to $H = 3.6\ m$, $T = 12.14\ s$ in full-scale). For Cases 7–11, the simulation cases applied to the current CFD model were combined with regular bow waves with the wave heights H = 0.032m, 0.048m, 0.064m, 0.080m, and 0.096m to evaluate the contribution of wave heights to the path-following performance. The wave heights considered are ranged from 2.4m to 7.2m in full-scale. The rationale for choosing the bow waves is that oblique waves have been identified as critical conditions that have a considerable impact on a ship's heading control, as stated in Kim et al. (2021c). It should be noted that the CFD analyses in this work were performed using deep water conditions.

2.2. Step 2: Modelling

The block diagram scheme of the path-following process adopted in

Table 1
The main properties of the KCS model used in this work.

Main particulars	Symbols	Model scale (1:75.24)	Full scale
Length between the perpendiculars	L_{BP}(m)	3.057	230.0
Length of waterline	L_{WL}(m)	3.0901	232.5
Beam at waterline	B_{WL}(m)	0.4280	32.2
Draft	D (m)	0.1435	10.8
Displacement	$\Delta(m^3)$	0.1222	52030
Block coefficient	C_B	0.651	0.651
Ship wetted area without rudder	S (m^2)	1.6834	9530
Longitudinal centre of buoyancy	% L_{BP}, fwd+	−1.48	−1.48
Metacentric height	GM (m)	0.008	0.6
Radius of gyration	K_{xx}/B	0.49	0.49
Radius of gyration	$K_{yy}/L_{BP}, K_{zz}/L_{BP}$	0.25	0.25
Propeller diameter	D_P(m)	0.105	7.9
Propeller rotation direction (view from stern)		Right hand side	Right hand side
Rudder turn rate	(deg/s)	20.1	2.32

this study is depicted in Fig. 4, in which the path-following control system for the ship free to move in 6 DOF is implemented using a PD controller (autopilot) in series with the LOS guidance algorithm. A series of waypoints should be first defined in the system to generate a feasible trajectory consisting of successive straight lines for path-following (*Step 2-1 Trajectory Generation*). Based on the generated trajectory and predicted ship position (from the free-running CFD model), the LOS guidance algorithm computes the desired yaw angle in order for the ship to successfully follow the pre-defined path (*Step 2-2 Guidance System*). The main function of the autopilot is to determine the rudder deflection angle, taking the desired yaw angle and the predicted yaw angle and rate as input (*Step 2–3 Control System*). The motion, location, and orientation of the ship in a given external environment are calculated by means of a dynamic ship model capable of resolving the complicated interactions between the hull, propeller, rudder and waves (*Step 2–4 Free-running CFD model*). The aforementioned features (from Step 2-1 to Step 2–4) should be properly modelled for the accurate prediction of the ship's path-following performance.

The present work adopted four different frames of reference for the simulation of path-following (Fig. 5), as can be found in the authors' previous work (Kim and Tezdogan, 2022): *(1) Earth-fixed coordinate ($o_o - x_o y_o z_o$), (2) Ship-fixed coordinate (o_s - $x_s y_s z_s$), 3) Propeller-fixed coordinate (o_p - $x_p y_p z_p$), and (4) Rudder-fixed coordinate (o_r - $x_r y_r z_r$).*

2.2.1. Step 2-1: Trajectory generation

The term "path-following" generally refers to the concept of following a predefined path independent of time. In this regard, the path that the vessel will follow should be first designed by defining a set of waypoints before conducting the path-following task. Fig. 6 depicts the predefined path designed for the path-following manoeuvre in this work and Table 3 presents the detailed path information consisting of waypoints, coordinates, true courses, and distances in model-scale. The

Table 2
The simulation cases to which the CFD model is applied in the model- and full-scales.

Model scale (1:75.24)							
Case no. C	Approach speed U_0 (m/s)	Propeller rev. (RPS)	Wave height H (m)	Encounter Angle μ (degrees)	Encounter Period T_e (s)	Wave steepness H/λ	Wave/ship length λ/L_{BP}
1	1.094	13.38	Calm water	–	–	–	–
2	0.860	13.38	0.048	180 (Head sea)	1.004	0.016	1.00
3	0.950	13.38	0.048	225 (Bow sea)	1.071	0.016	1.00
4	1.070	13.38	0.048	270 (Beam sea)	1.399	0.016	1.00
5	1.030	13.38	0.048	315 (Quartering sea)	2.103	0.016	1.00
6	1.050	13.38	0.048	0 (Following sea)	2.694	0.016	1.00
7	1.004	13.38	0.032	225 (Bow sea)	1.055	0.010	1.00
8	0.950	13.38	0.048	225 (Bow sea)	1.071	0.016	1.00
9	0.876	13.38	0.064	225 (Bow sea)	1.087	0.021	1.00
10	0.811	13.38	0.080	225 (Bow sea)	1.104	0.026	1.00
11	0.737	13.38	0.096	225 (Bow sea)	1.123	0.031	1.00
Full scale							
Case no. C	Approach speed U_0 (m/s)	Propeller rev. (RPS)	Wave height H (m)	Encounter Angle μ (degrees)	Encounter Period T_e (s)	Wave steepness H/λ	Wave/ship length λ/L_{BP}
1	9.489 (18.4 knots)	1.54	Calm water	–	–	–	–
2	7.459 (14.5 knots)	1.54	3.6	180 (Head sea)	8.709	0.016	1.00
3	8.240 (16.0 knots)	1.54	3.6	225 (Bow sea)	9.289	0.016	1.00
4	9.281 (18.0 knots)	1.54	3.6	270 (Beam sea)	12.135	0.016	1.00
5	8.934 (17.4 knots)	1.54	3.6	315 (Quartering sea)	18.241	0.016	1.00
6	9.107 (17.7 knots)	1.54	3.6	0 (Following sea)	23.368	0.016	1.00
7	8.781 (16.9 knots)	1.54	2.4	225 (Bow sea)	9.151	0.010	1.00
8	8.240 (16.0 knots)	1.54	3.6	225 (Bow sea)	9.289	0.016	1.00
9	7.598 (14.8 knots)	1.54	4.8	225 (Bow sea)	9.429	0.021	1.00
10	7.035 (13.7 knots)	1.54	6.0	225 (Bow sea)	9.576	0.026	1.00
11	6.393 (12.4 knots)	1.54	7.2	225 (Bow sea)	9.741	0.031	1.00

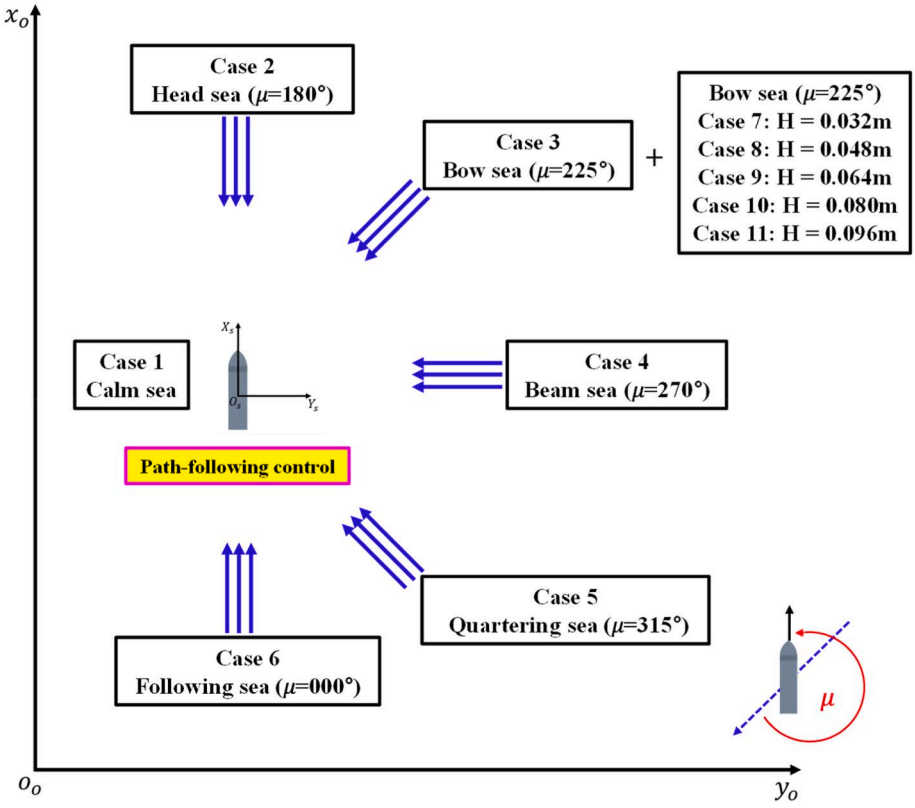


Fig. 3. The Schematic view of the simulation cases applied to this study.

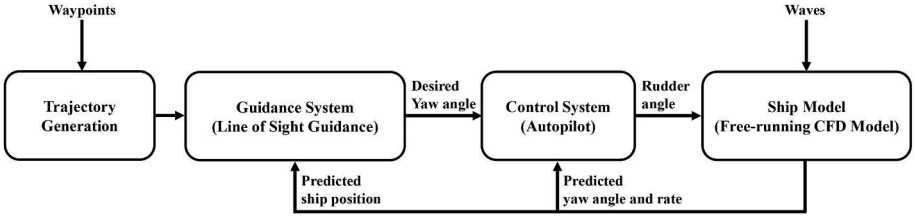


Fig. 4. Block diagram of the path-following control system in this study.

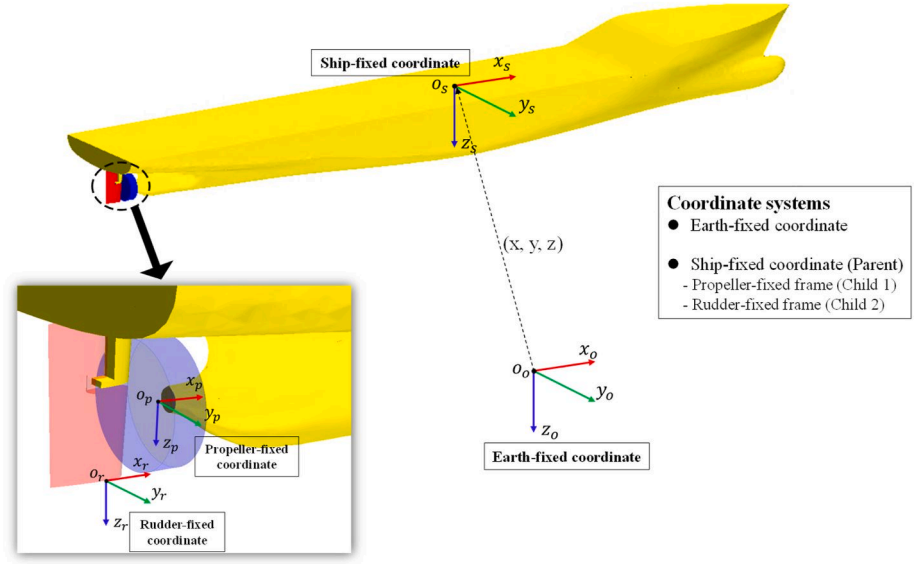


Fig. 5. The four reference coordinate systems used in this study, adapted from Kim and Tezdogan (2022).

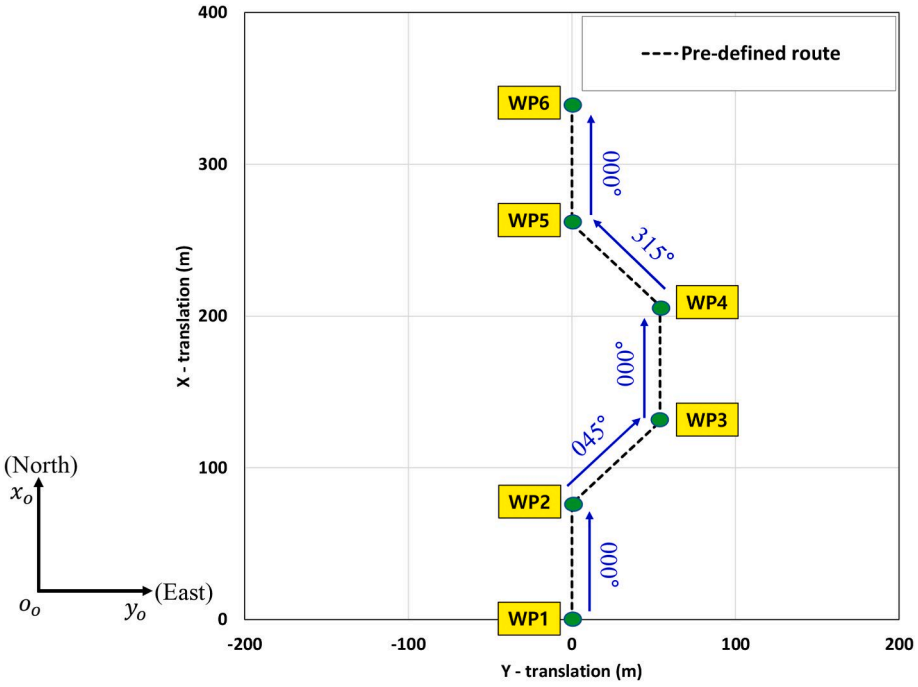


Fig. 6. The generated route for the path-following manoeuvre in this work.

Table 3
Passage plan from Waypoint 1 to Waypoint 6.

Waypoint no.	Coordinate (x,y)	True course (°)	Distance (m)
1	(0,0)	000°	76.4 (25 L_{BP})
2	(76.4,0)	045°	76.4 (25 L_{BP})
3	(130.4, 54)	000°	76.4 (25 L_{BP})
4	(206.8, 54)	315°	76.4 (25 L_{BP})
5	(260.8, 0)	000°	76.4 (25 L_{BP})
6	(337.2, 0)	–	–

prescribed path was formed in order to simply evaluate the performance of the straight-line path following control and the course alteration to the starboard/port side during the path-following manoeuvre in a given external environment. It is also important to note that the pre-defined path comprised successive straight-line segments (the path-tangential angle is constant for each line), each of which is indicated by the true course between two consecutive waypoints. As presented previously (Fig. 4), the predefined path is intended to be fed into the LOS guidance algorithm as input to compute the desired yaw angle which forces the ship to follow the path.

2.2.2. Step 2-2: Guidance system

This sub-section deals with the Line-of-Sight guidance law adopted in this study for the horizontal plane guidance of a surface vessel performing path-following (Fossen et al., 2003). It has to be stressed that the LOS guidance law computes the heading angle command (i.e., the desired heading angle) to the autopilot in order for the ship to converge to the predefined path on the x_oy_o plane. The geometry of the LOS guidance law for straight lines is presented in Fig. 7, along with some of

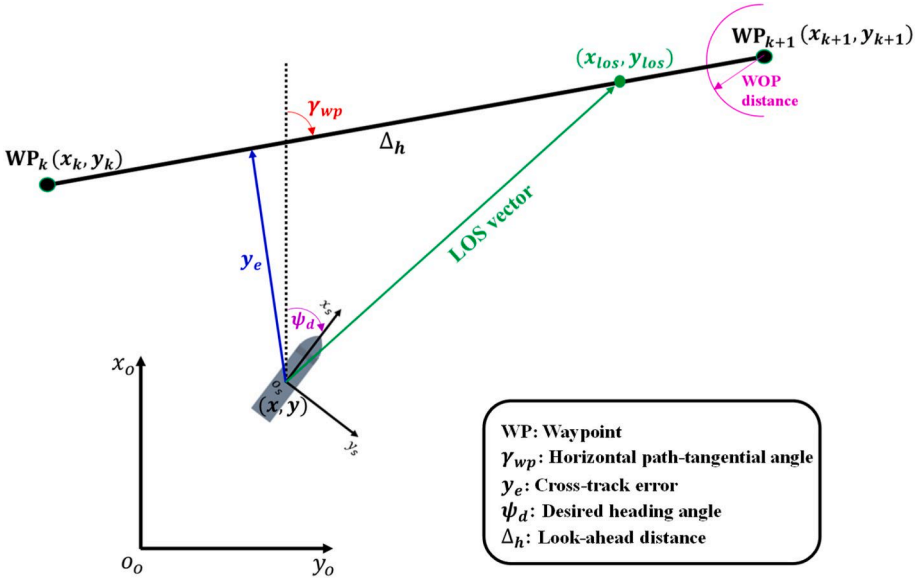


Fig. 7. The illustration of the line-of-sight guidance law for straight lines in the x_oy_o plane.

the main variables involved in the LOS algorithm. The desired heading (Ψ_d) formulation used in the LOS algorithm can be expressed as follows:

$$\Psi_d = \gamma_{wp} + \arctan\left(\frac{-y_e}{\Delta_h}\right) - \beta \tag{1}$$

$$\gamma_{wp} = \text{atan } 2(y_{k+1} - y_k, x_{k+1} - x_k) \tag{2}$$

$$y_e = -(x - x_k)\sin(\gamma_{wp}) + (y - y_k)\cos(\gamma_{wp}) \tag{3}$$

$$\beta = \text{atan } 2(v, u) \tag{4}$$

in which, γ_{wp} represents the horizontal path-tangential angle. The two-argument function atan2(y, x) returns the arctangent of the two numbers x and y, considering the signs of both x and y to determine the quadrant of the results. (x_k, y_k) and (x, y) indicate the position of the k-th waypoint and the position of the ship's centre of gravity (CG), respectively (expressed in the earth-fixed system of reference). y_e is the cross-track error for a given ship position (x, y). Δ_h denotes the steering law turning parameter which is called the lookahead distance. According to Fossen (2011), the lookahead distance (Δ_h) is given in metres and is typically defined between 1.5 and 2.5 of a ship length L_{BP}. It is noteworthy to mention that reaching the desired path can be faster achieved by applying a small lookahead distance, inducing more aggressive steering. On the other hand, a large lookahead distance will induce smoother steering, hence resulting in reaching the desired path slower. In this study, a constant Δ_h of 2.0 L_{BP} was used with an aim to compare the path-following performance more reliably in different wave conditions by reducing the overall complexity of the problem. β is the drift angle experienced by the ship during the path-following manoeuvre. It has been observed in simulations and real operations that a ship may deviate from the predefined path while changing its course towards the next waypoint, mainly stemming from a late course change. To reduce such deviations occurring while changing course, Wheel Over Point (WOP), which represents a point where a ship starts to change the course towards the next waypoint, should be determined. For simplicity, the ship was forced to track the next waypoint (WP_{k+2}) when the ship is inside a radius of 2.0 L_{BP} (WOP distance) from the current tracking waypoint (WP_{k+1}) throughout all the simulations, as shown in Fig. 7. The LOS control objective for the horizontal plane guidance of the ship performing path-following is to minimise the cross-track error (y_e), which can be given by:

$$\lim_{t \to +\infty} y_e(t) = 0. \tag{5}$$

2.2.3. Step 2–3: Control system

In this work, a PD control system was used to generate the necessary rudder action to achieve the path-following task. The control function is given as follows:

$$\delta(t) = K_p e(t) + K_d \frac{de(t)}{dt} \tag{6}$$

$$e(t) = \psi(t) - \psi_d \tag{7}$$

where t indicating the time (s) elapsed after the start of the path-following manoeuvre, $\delta(t)$ the rudder angle (°), $\psi(t)$ the instantaneous yaw angle (°) at a given time, ψ_d the desired yaw angle (°) which was determined from the LOS algorithm. K_p and K_d represent the proportional and derivative control gains, respectively. For the CFD model in this study, the control gains were determined by the trial-and-error method ($K_p = 5$ *and* $K_d = 10$). As seen in Fig. 8, the PD controller computes the necessary rudder deflection angle with the consideration of the difference between the instantaneous yaw angle and the desired yaw angle to force the ship to follow the predefined path.

2.2.4. Step 2–4: Free-running CFD model

The free-running CFD simulations in this paper were carried out using the commercial CFD code STAR-CCM+ (version 15.04) developed by Siemens PLM Software. This sub-section provides a detailed description of the numerical scheme employed in this work.

The CFD code solves the unsteady Reynolds Averaged Navier-Stokes (URANS) equations for turbulent flows around the ship, under the assumption of Newtonian, incompressible, and viscous fluid. Without taking into account the body forces, the following vectorial form can be used to express the continuity and momentum equations for the flow:

$$\nabla \bullet \mathbf{U} = 0 \tag{8}$$

$$\frac{\partial(\rho \mathbf{U})}{\partial t} + \nabla \bullet \left[\rho(\mathbf{U} - \mathbf{U}_g)\mathbf{U}\right] = -\nabla p + \nabla \bullet \left(\mu_{\text{eff}} \nabla \mathbf{U}\right) + \nabla \mathbf{U} \bullet \nabla \mu_{\text{eff}} + q_i \tag{9}$$

where $\nabla\bullet$ $(\bullet)$ and ∇ $(\bullet)$ represent the divergence and gradient operators, respectively. $\mathbf{U}$ is the fluid velocity and $\mathbf{U}_g$ is the grid velocity; p is the static pressure; ρ is the fluid density; $\mu_{\text{eff}} = \rho(\nu + \nu_t)$ indicates the effective dynamic viscosity, where ν and ν_t stand for the kinematic and eddy viscosity, respectively (ν_t is obtained from the turbulence model); q_i is a user-defined source term.

The turbulence model employed in this work was the Shear Stress Transport (SST) model for the closure of the governing equations, which was developed by Menter (1994). The SST turbulence model combines the benefits of the k-ε turbulence model in the far-field and the k−ω turbulence model in the near-wall region. As one of the most common two-equation RANS models, the Menter's SST model has been widely applied in free-running CFD simulations in the open literature, presenting promising results (Mofidi and Carrica, 2014; Wang and Wan, 2018; Wang et al., 2018; Kim et al., 2021c, 2022c).

Three-dimensional free-surface flows around the ship were modelled utilising the Volume of Fluid (VOF) approach, and the transport equation can be expressed as (Siemens, 2020):

$$\frac{\partial \alpha_i}{\partial t} + \nabla \bullet \left[\alpha_i(\mathbf{U} - \mathbf{U}_g)\right] = S_{\alpha_i} - \frac{\alpha_i}{\rho_i}\frac{D\rho_i}{Dt} - \frac{1}{\rho_i}\nabla \bullet \left(\alpha_i \rho_i \mathbf{v}_{d,i}\right) \tag{10}$$

where α_i indicates the volume fraction of phase i and its value varies from 0 to 1 to describe the relative proportion of fluid in each cell ($\alpha = 0$: air, $0 < \alpha < 1$: two-phase interface, $\alpha = 1$: water). S_{α_i} is a user-defined source term of phase i; $\frac{D\rho_i}{Dt}$ is the Lagrangian derivative of the phase densities ρ_i; $\mathbf{v}_{d,i}$ is the diffusion velocity.

The RANS-VOF solver used in this study employed a finite volume method to discretise the governing partial differential equations into a set of linear algebraic equations. The temporal discretisation of the transient terms was based on a second-order implicit backward Euler

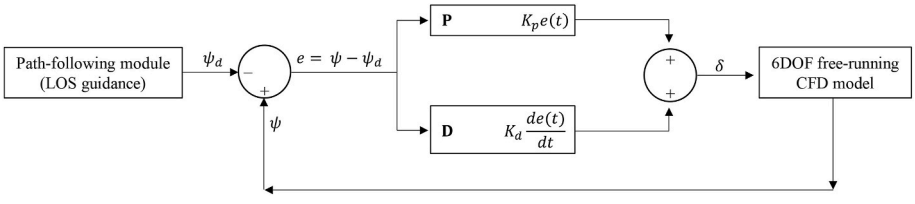
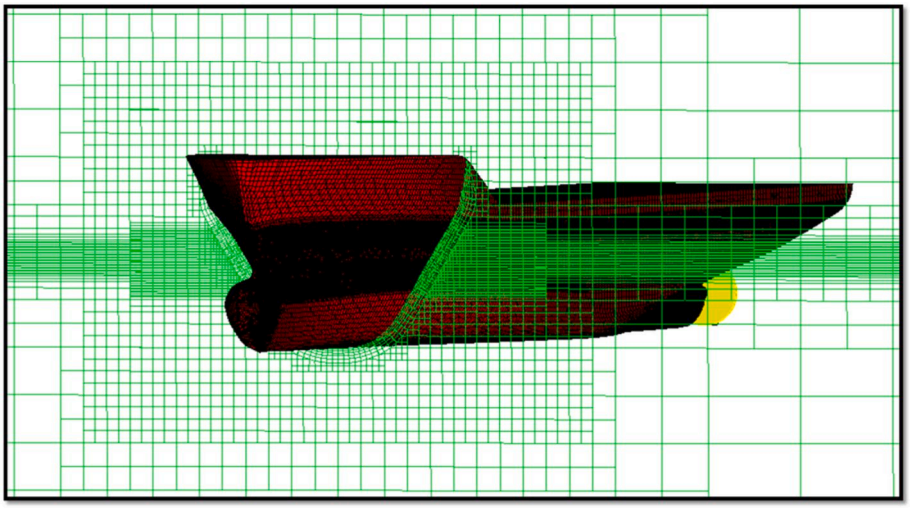


Fig. 8. The block diagram of the autopilot applied to the CFD model for the path-following.

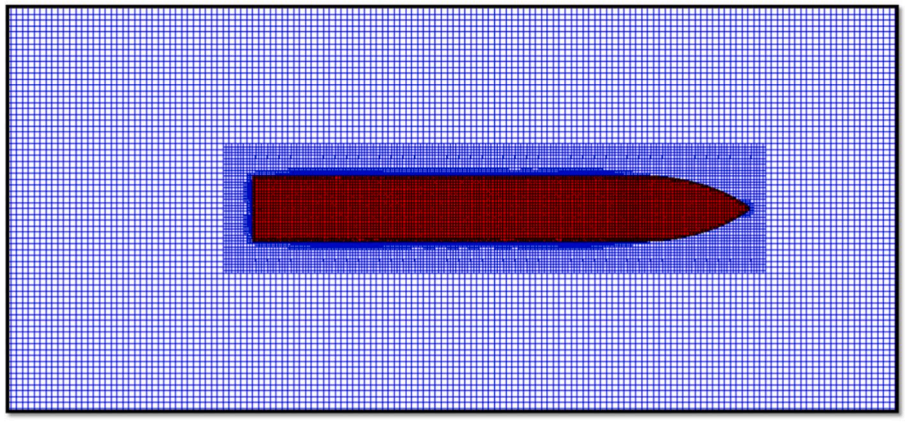
scheme. The convection terms and diffusive terms were discretised with a second-order upwind scheme and a second-order centred scheme, respectively. A SIMPLE-type algorithm was adopted for the treatment of the pressure-velocity coupling.

The current CFD model simulated the average effect of the KCS propeller (KP505) operating in the behind-hull conditions, using an actuator disk model based on the body-force method. The main features of the body-force method adopted in this work are discussed briefly here. More detailed information can be found in Kim et al. (2021b); Kim et al. (2022a). The body-force approach distributes a field of axial and tangential body forces within a disk of finite thickness to account for the presence of the propeller behind the hull of a ship, simulating the acceleration and the increase in the swirl experienced by the flow as it passes through the propeller (Broglia et al., 2013). The main benefit of this method is its capability to provide a reliable estimation of the flow field caused by the complicated interactions between the hull, the propeller, and the rudder, while also being computationally efficient. In order to complete the propeller modelling, some input parameters, such as the open water results for the KP505 propeller, the geometric properties, and the propeller revolution speed to be used in the simulation, are required to be entered into the CFD software (Siemens, 2020). In the authors' previous papers (Kim et al., 2021c, 2022c), the actuator disk model for the KCS propeller has exhibited promising results in predicting the manoeuvring behaviour of the ship performing a zigzag manoeuvre in shallow water and turning circle manoeuvres in both calm water and head seas.

The computational grids for the KCS appended with the rudder were discretised using the built-in meshing capability of STAR-CCM + which employs the Cartesian cut-cell method (Song et al., 2020). For the volume mesh generation, a trimmed cell mesher was chosen due to its capability to generate a high-quality grid for complex mesh generating problems such as free-surface interactions with bodies. The surface grid was produced with the use of a surface remesher, which is a robust approach to optimise the geometry surface for the volume meshes (Siemens, 2020). Orthogonal prismatic cells next to wall surface boundaries were generated utilising a prism layer mesh model to capture the turbulent boundary layer close to the hull. Several grid refinements were made to better resolve the flow features in the areas around the bow, stern, the tight gap parts between the rudder blade and horn, and the free surface region. It is worth mentioning that 20 grid points per wave height in the z direction and 80 grid points per wavelength in the x and y directions were generated to ensure stable wave propagation, in accordance with the guidelines for ship CFD applications from ITTC (2011). A dynamic overlapping grid approach was used to generate proper moving grids over the moving ship and the deflecting rudder surface during path-following manoeuvres, resulting in the computational domain divided into three different regions: 1) background region, 2) hull overset region, and 3) rudder blade overset



(a) Frontal view



(b) Top view

Fig. 9. The computational mesh for the present CFD model.

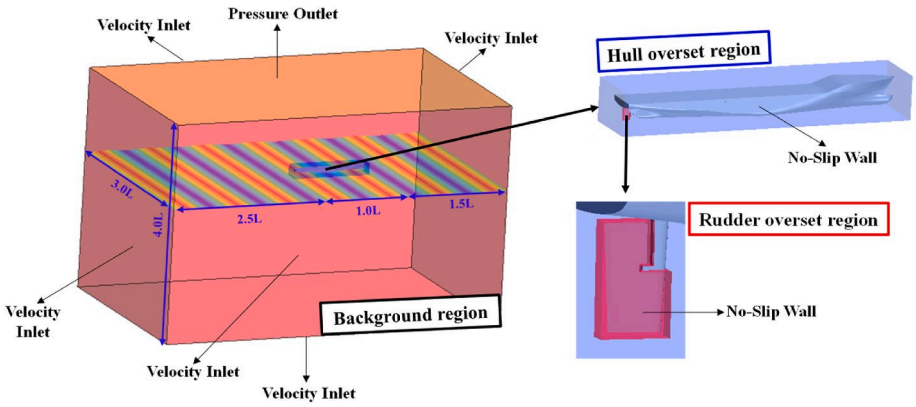


Fig. 10. A general view of the background and overset regions and the imposed boundary conditions.

region. Fig. 9 displays an overview of the final computational domain mesh for the ship path-following simulations. The total grid number generated in the calm water case (Case 1) and wave simulations (Cases 2–11) is approximately 5 and 8 million cells, respectively. A general view of the computational domain with the imposed boundary conditions is depicted in Fig. 10. For a detailed description of the applied boundary condition for the current CFD model, reference can be made to the authors' previous studies (Kim et al., 2021c, 2022a). In this work, the regular waves were generated using a fifth-order Stokes wave model capable of incorporating the nonlinearities of the waves, based on the theory developed by Fenton (1985). The time step size was set as 0.005s throughout all the simulations, which complies with the guideline from ITTC (2011) that at least 100 time steps per encounter period for regular wave simulations should be used for desired wave propagation during the simulations.

2.3. Step 3: Results of analysis

Step 3 was organised to implement the free-running CFD model developed in line with the modelling setup presented in Step 2 for the path-following applications in regular waves. This task is to demonstrate the effectiveness of the CFD model in predicting the path-following capability of a ship as well as to comprehensively investigate the effects of waves on the path-following behaviour. The performance of the ship to follow a predefined path was discussed in terms of the LOS guidance parameters, trajectory, and kinematic parameters, which can provide a deeper understanding of the path-following performance of a ship for autonomous navigation in a real seaway.

3. Results

3.1. Validation and verification study

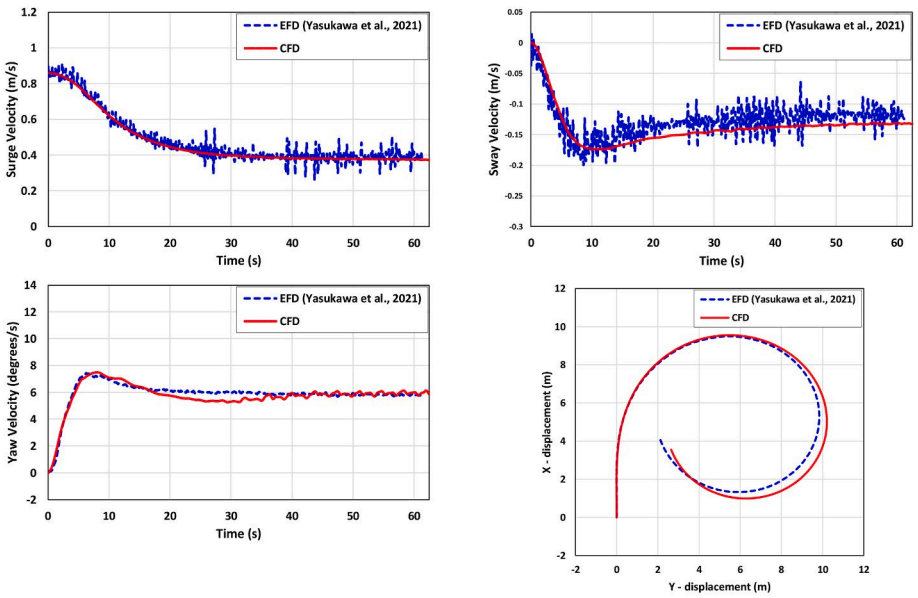
The accurate prediction of the path-following performance is highly dependent on the capability of the model (to be used for the path-following simulations) in estimating the ship trajectory changing in response to the rudder deflection. Thus, it is of critical importance to first validate the reliability of the model in predicting the manoeuvring performance of the ship within an acceptable level of accuracy, with particular attention to the trajectory in response to the rudder behaviour. Free-running model tests for the KCS model adopted in this work were carried out by Hiroshima University (Yasukawa et al., 2021), consisting of turning circle manoeuvres in calm water and regular head seas. In the authors' previous study (Kim et al., 2021c), the trajectory and the time histories of the kinematic parameters (such as surge, sway, yaw velocities) predicted by the free-running CFD model during the manoeuvres were presented and compared with the experimental data from the free-running tests. Fig. 11 illustrates some of the comparison results provided in Kim et al. (2021c). The comparison in terms of the ship trajectory and kinematic parameters during the manoeuvre confirmed good agreement between the free-running test (Yasukawa et al., 2021) and the CFD results. Given this, it can be claimed that the free-running CFD model adopted in this work can accurately capture the complex fluid-structure interactions between the hull, rudder, propeller, and waves, resulting in the reliable prediction of the ship trajectory in response to the rudder deflection. Therefore, it is thought that the current CFD model using the LOS guidance law is sufficiently reliable for tackling the path-following problem in question.

In addition, in Kim et al. (2021c)'s study, a spatial and temporal convergence study was carried out to quantify the numerical uncertainties of the CFD model using the same numerical scheme used in this work based on the Grid Convergence Index (GCI) method. It is reported that the CFD uncertainties for the critical manoeuvring indices (closely related to the ship trajectory) were predicted to be a maximum of 0.28% in the grid sensitivity study and 0.19% in the time-step sensitivity study. For any further details regarding the validation and verification studies performed, reference can be made to Kim et al. (2021c).

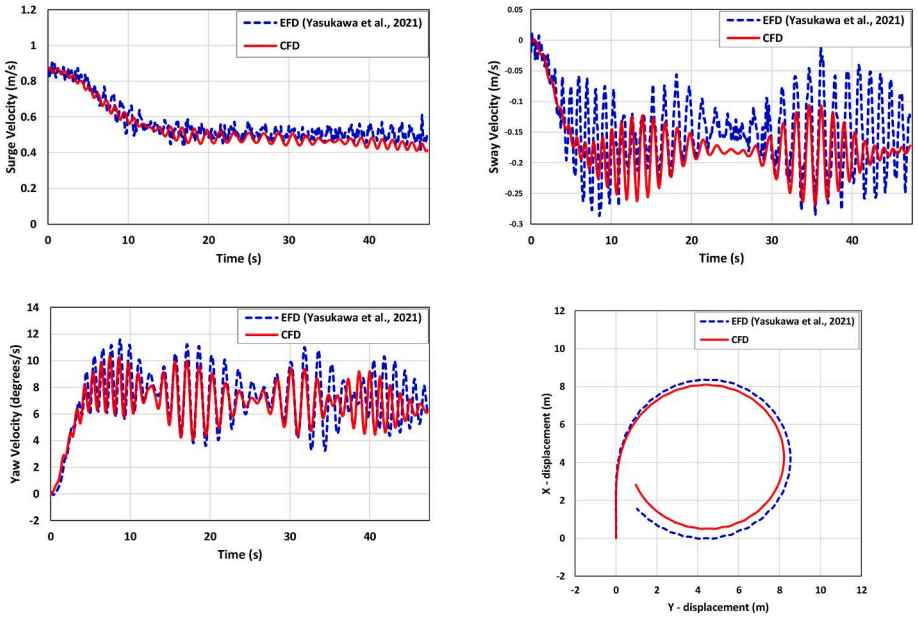
3.2. Path-following manoeuvre in calm water (case 1)

The CFD results obtained for the path-following manoeuvre in calm deep water (Case 1) are described in this sub-section. Fig. 12 displays the *xy*-plot of the predicted ship trajectory (black dashed line) during the path-following control, together with the predefined geometrical route (red solid line) consisting of five successive straight line segments. In the figure, the position of the ship is referred to the earth-fixed system of reference, and the origin point (0,0) is the location at which the path-following manoeuvre started based on the two interconnected systems (i.e., the LOS guidance system and the heading autopilot). It was seen that the ship followed the predefined path very well in calm water, showing a good performance of the straight-line path following control as well as the course alteration to the starboard/port side, clearly evidenced in Fig. 12. It is worth noting that the application of WOP enabled the manoeuvring ship to reduce a deviation from the predefined path while changing its course towards the next waypoint. Without this WOP application, relatively large deviations from the prescribed route would have been observed near the waypoints 2, 3, 4, and 5, resulting from a late course change.

For the sake of completeness, the time histories of the critical LOS guidance parameters, kinematic parameters, and ship motions observed during the path-following manoeuvre in calm water are reported in Fig. 13. As Fig. 13 (a) and (b) jointly show, the ship's heading angle was observed to successfully converge to the desired heading angle determined by the LOS guidance system (Eq. (1)), demonstrating the effectiveness of the applied autopilot system (the PD controller) for path-



(a) Calm water



(b) Regular head seas

Fig. 11. The comparison of the trajectory and kinematic parameters experienced by the ship performing turning manoeuvres in calm water and regular head seas, adapted from Kim et al. (2021c).

following in calm water. The reason for the slight difference between the desired and predicted heading angle is that the ship cannot turn instantly to the desired heading angle, mainly due to the yaw inertia related to the size of the ship and the control gain gains used in the PD controller. It can also be observed from Fig. 13 (b) that the rudder blade was slightly deflected to the port side (i.e., the positive sign of the rudder angle) when the ship was advancing forward. It can be explained by the asymmetric pressure distribution between the starboard and the port of the rudder blade, because of a non-uniform flow exerted by the right-handed propeller, as discussed in depth in Kinaci et al. (2021); Kim et al. (2022a). This resulted in a small rudder normal force, and thus caused the yaw moment to turn the ship's heading to the starboard to a small extent, consequently leading to a counter rudder angle (the positive rudder angle) to make the ship advance straight. The abrupt change of the rudder angle that occurred at around 64, 138, 212, and 286 s was a corrective action against sudden changes in the path-tangential angle (Fig. 13 (c)) and cross-track error (Fig. 13 (d)) when the ship was within a radius of the WOP distance from the current tracking waypoint. It has to be stressed that the positive value of the cross-track error (y_e) indicates that the ship performing the path-following manoeuvre is positioned on the starboard side of the predefined path. Eq. (1) is designed in a way such that the positive y_e results in the negative value of $\arctan\left(\frac{-y_e}{\Delta_h}\right)$, hence the desired heading angle becomes smaller than the path-tangential angle (γ_{wp}). This desired heading angle can ensure the positional convergence to the reference path. The negative value of the cross-track error, on the other hand, means that the ship is positioned port side of the predefined path, leading to the positive value of $\arctan\left(\frac{-y_e}{\Delta_h}\right)$ (Ψ_d becomes greater than γ_{wp}).

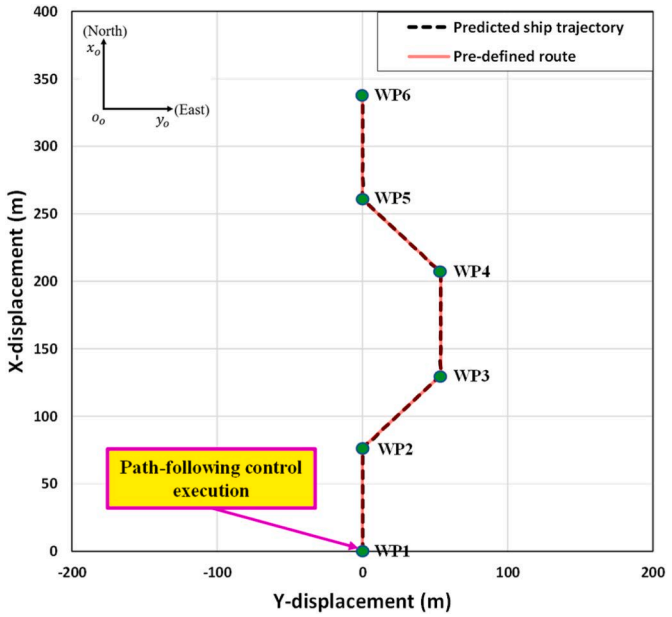


Fig. 12. *xy*-plot of the predefined and predicted ship path during the path-following manoeuvre in calm deep water (Case 1).

It was observed that the ship suffered involuntary surge speed losses while changing its course towards the next waypoint, and experienced speed recoveries after the completion of the course alteration (Fig. 13 (f)). A close inspection to Fig. 13 (b) and (e) revealed that such speed changes were closely correlated with a change in the ship resistance caused by the rudder resistance and the drift angle experienced by the ship during the manoeuvre. When the ship passed the WOP established on the predefined path, the rudder started to be deflected in order for the ship's heading to converge to the desired heading angle determined by the LOS guidance law. The deflected rudder blade generated a relatively large sway force and yaw moment to make the ship turn, resulting in changes in the sway and yaw velocities (as seen in Fig. 13 (g) and (h)). As expected in the absence of external disturbances, only small heave and pitch motions were noticed during the path-following manoeuvre in calm water (Fig. 13 (i) and (j)). Every time the ship changed its course towards the next waypoint, a relatively large roll motion occurred, attributed from the strong rudder normal force induced by the deflected rudder (evidenced in Fig. 13 (k)). The maximum roll angle experienced by the ship performing the path-following in calm water remained below 6°. The global wave pattern around the ship during the path-following manoeuvre in calm water is shown in Fig. 14, in which the Kelvin wave generated by the manoeuvring ship is clearly visible.

3.3. *Path-following manoeuvres in waves of different directions (cases 2–6)*

This sub-section will present the path-following performance of the ship operating in waves of different propagation directions (Cases 2–6). It has to be stated that the path-following manoeuvres in the waves started from the stable state of the heading control condition in which the desired yaw angle (described in Eq. (7)) was set to be 0°. In Fig. 15, the *xy*-plot of the predicted trajectories experienced by the ship performing the path-following control was presented, together with the reference trajectory (red solid line) comprised of the five straight lines interconnecting the waypoints. The three zoomed-in views of the ship trajectories near the waypoints are also depicted in the figure for clarity. It can be observed that the path-following task under different wave direction conditions was successfully performed based on the LOS guidance law, presenting the predicted ship paths almost consistent with the predefined route from a practical point of view. It is worth noting that the contribution of wave directions to the ship trajectories under path-following control is evidenced in the zoomed-in images, in terms of deviations from the planned route in comparison with the inherent path-following performance in calm water. When the ship performed the path-following manoeuvre with true course 000° (e.g., WP1-WP2, WP3-WP4, and WP5-WP6), the beam and quartering seas (Cases 4 and 5) caused the greatest oscillatory deviation from the predefined path. This maximum deviation estimated at the stabilised phase was seen to have an amplitude of approximately 0.5m, which corresponds to 0.16 L_{BP}. The main reason for such a deviation can be explained by the asymmetric pressure distribution on the ship hull caused by the incident waves. A good discussion of the wave direction effects on the pressure distribution around the ship hull is described in Kim et al. (2021c), who carried out free-running CFD simulations in waves of different wave directions for the KCS. Path deviations in the bow seas (Case 3) were noted to some extent for the same reason. As expected, negligible path deviations were observed in the head and following seas (Cases 1 and 6) due to the symmetric pressure distribution around the ship hull. It is worth noting that the ship experienced successive changes in the wave-encounter direction when performing the path-following control along the reference straight lines, attributed to consecutive variations in the true course. For example, the ship operating in the head seas (Case 2) encountered the head seas (WP1-WP2), port bow seas (WP2-WP3), head seas (WP3-WP4), starboard bow seas (WP4-WP5), and head seas (WP5-WP6) in series after starting the path-following manoeuvre. It can be also observed that when the ship changed its course to track the next waypoint, relatively large path deviations occurred near the waypoint in some cases (for example, WP3: Cases 5 and 6, WP5: Cases 4 and 6). This exemplifies the challenge of the course change when the incident wave is coming from the ship's beam and quarter and its encounter direction is similar to the intended turning direction. It should be noted that by increasing the WOP distance, the ship's path could be less deviated near the waypoint when altering course towards the next waypoint.

In Fig. 16, the time histories of the critical parameters concerning the LOS guidance law, kinematic parameters, and ship motions predicted during the path-following control in waves of different wave directions are presented, where colours are chosen consistent with those in Fig. 15. It is worth noting that throughout all the cases (Case 2–6) in this sub-section, the ship's heading angle successfully converged to the desired heading angle. Relatively larger fluctuation amplitudes of the desired and predicted yaw angle occurred when the ship experienced the oblique and beam seas, coherently with the trend of the path deviation discussed in Fig. 15. Such oscillatory behaviours in the desired yaw angle are closely correlated with the contribution of the wave propagation direction to the cross-track error which should be minimised to achieve the LOS control objective. As expected, the rudder deflection angle also became remarkably oscillatory when the ship encountered the oblique and beam seas in order to make the ship's heading angle converge to the target one. In addition, the oblique and beam seas caused a relatively larger rudder deflection angle to keep the ship following the straight-line path, as seen in the graphs for the rudder angle. Similar to the calm water case (Case 1), the head and following seas led to a small fluctuation in the ship's heading and rudder angles while performing the path-following task. Immediate changes in the path-tangential angle and cross-track error were noted when the ship passed the WOP. As stated previously, large fluctuations of the cross-track error in the oblique and quartering seas were intimately related to the oscillatory desired yaw angle observed during the path-following manoeuvre, as illustrated in Eq. (1). It can be seen that the ship altered its course towards the next waypoint with a relatively larger drift angle compared to the calm water case. According to the graphs for the drift angle, CFD calculations imply that the ship encountering the quartering sea experienced the maximum drift angle (about 25°).

Inspecting the graphs for the surge velocity during the path-following control, there was a strong correlation between speed recoveries (after a sharp speed loss) and the wave-encounter directions. It

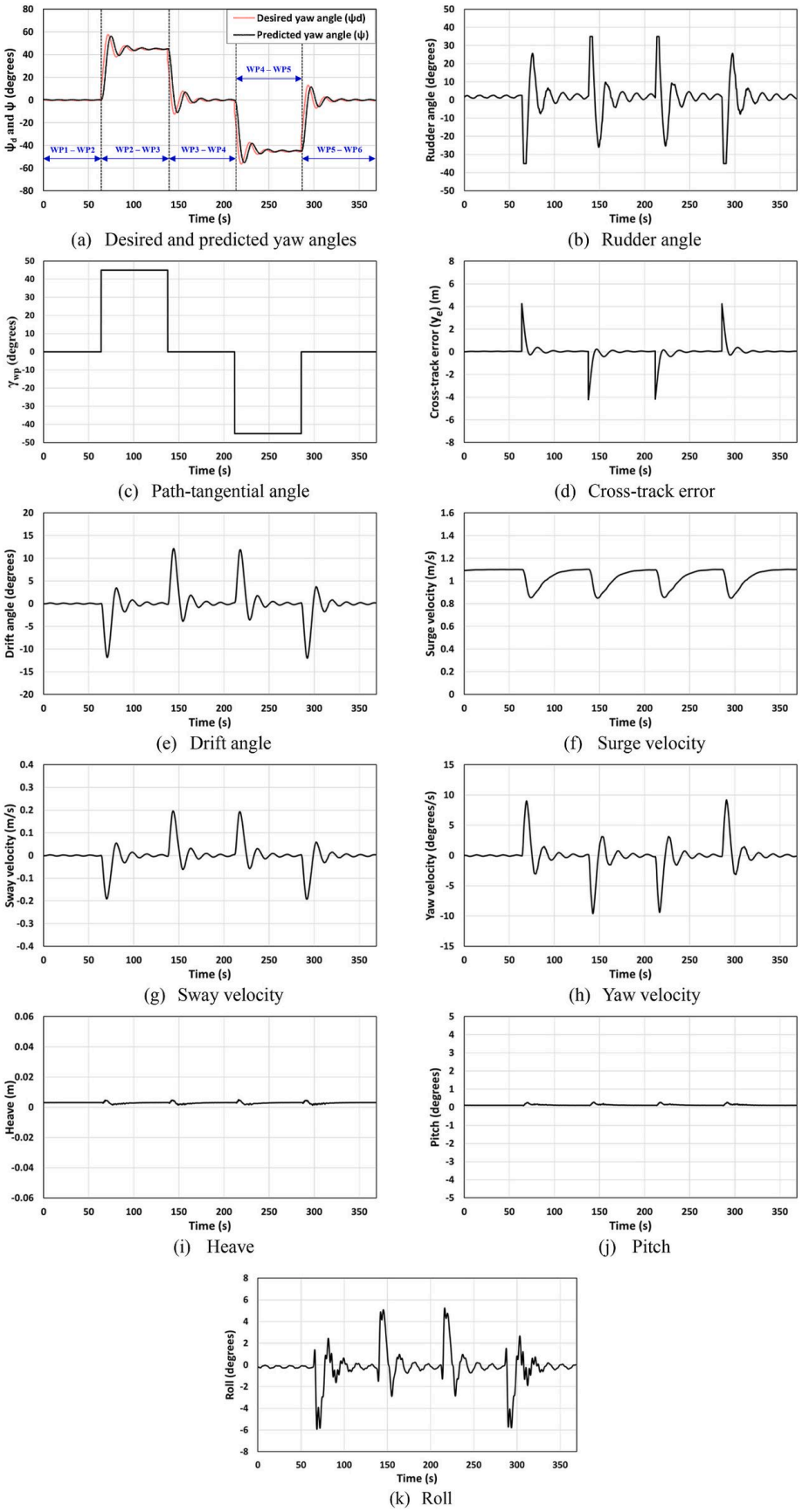


Fig. 13. The time histories of the critical LOS guidance parameters, kinematic parameters, and ship motions during the path-following manoeuvre in calm water (Case 1).

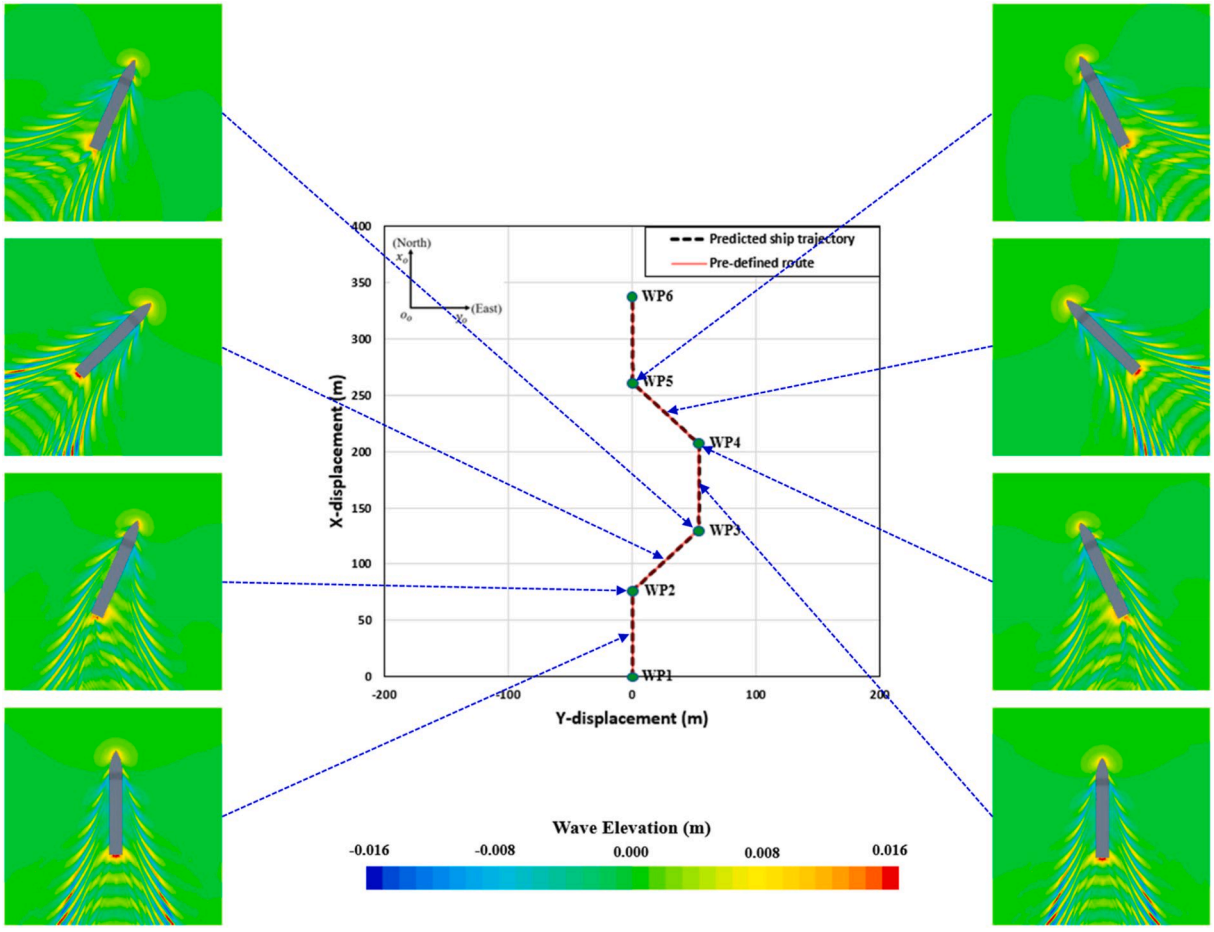


Fig. 14. Measured wave patterns around the ship performing the path-following manoeuvres (Case 1).

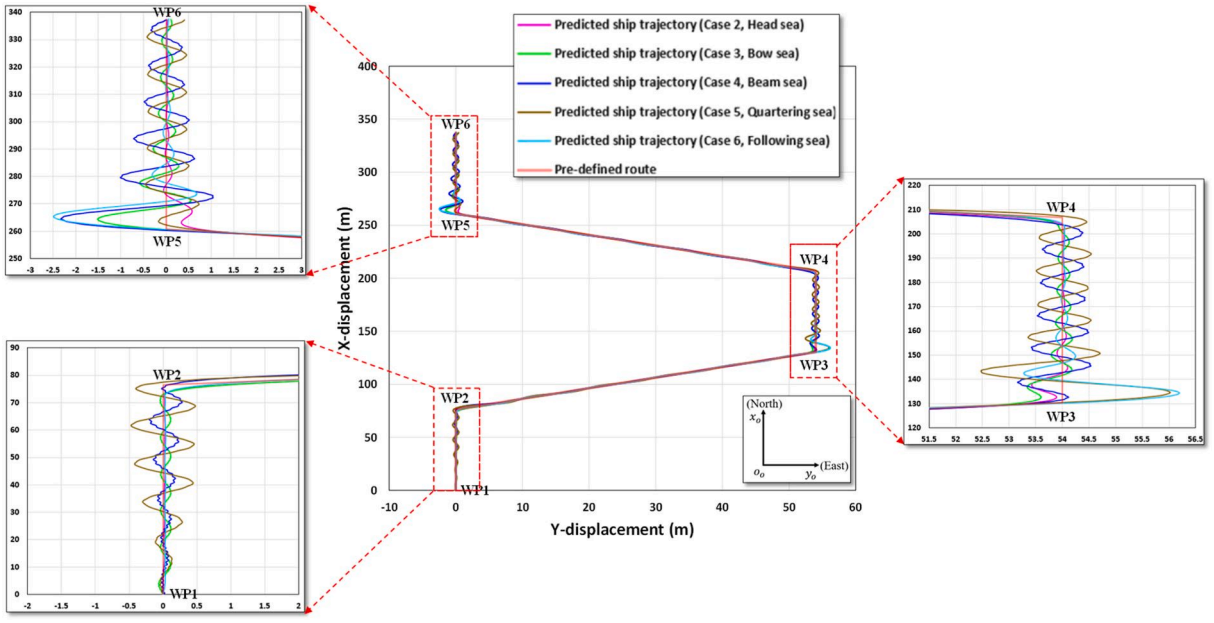
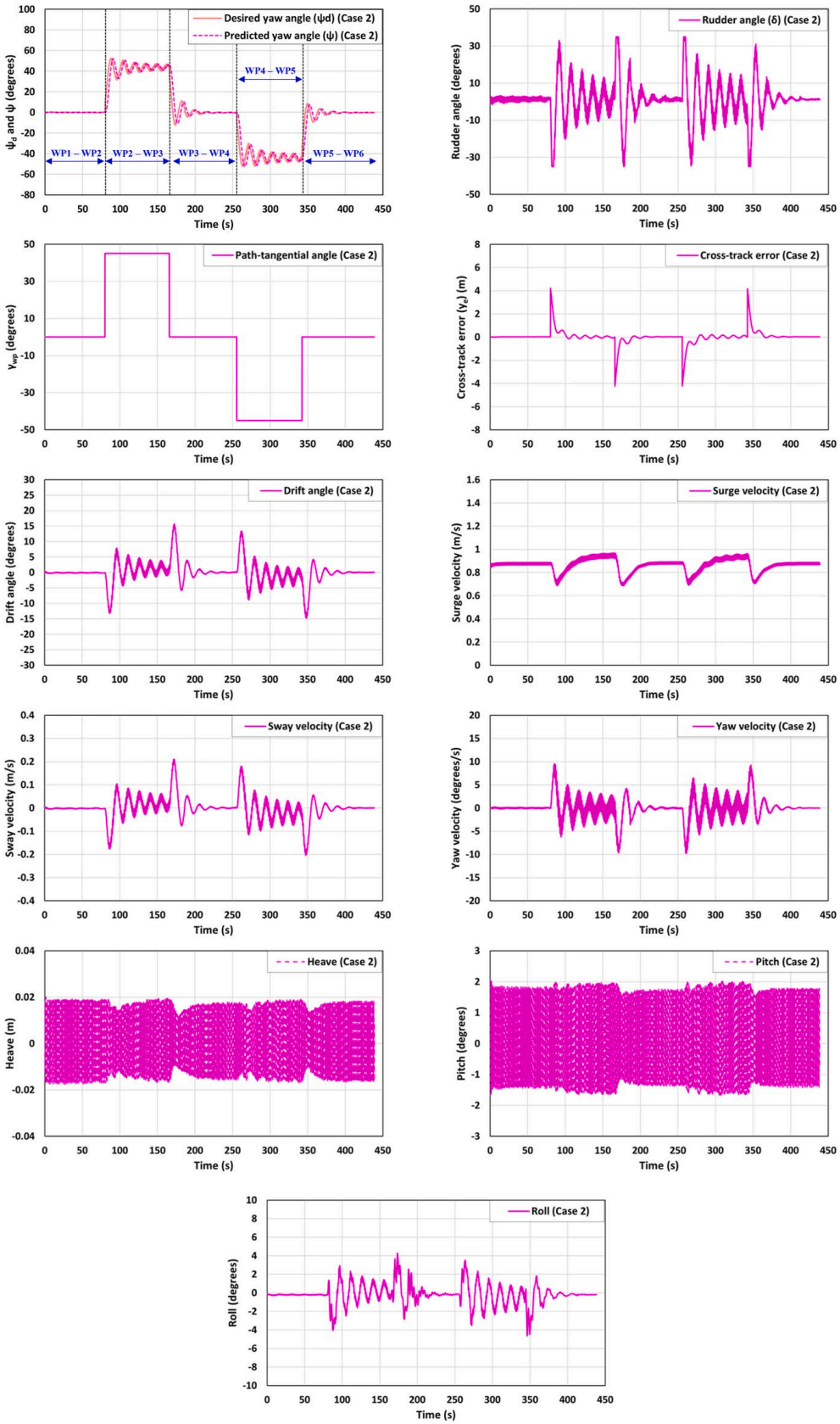
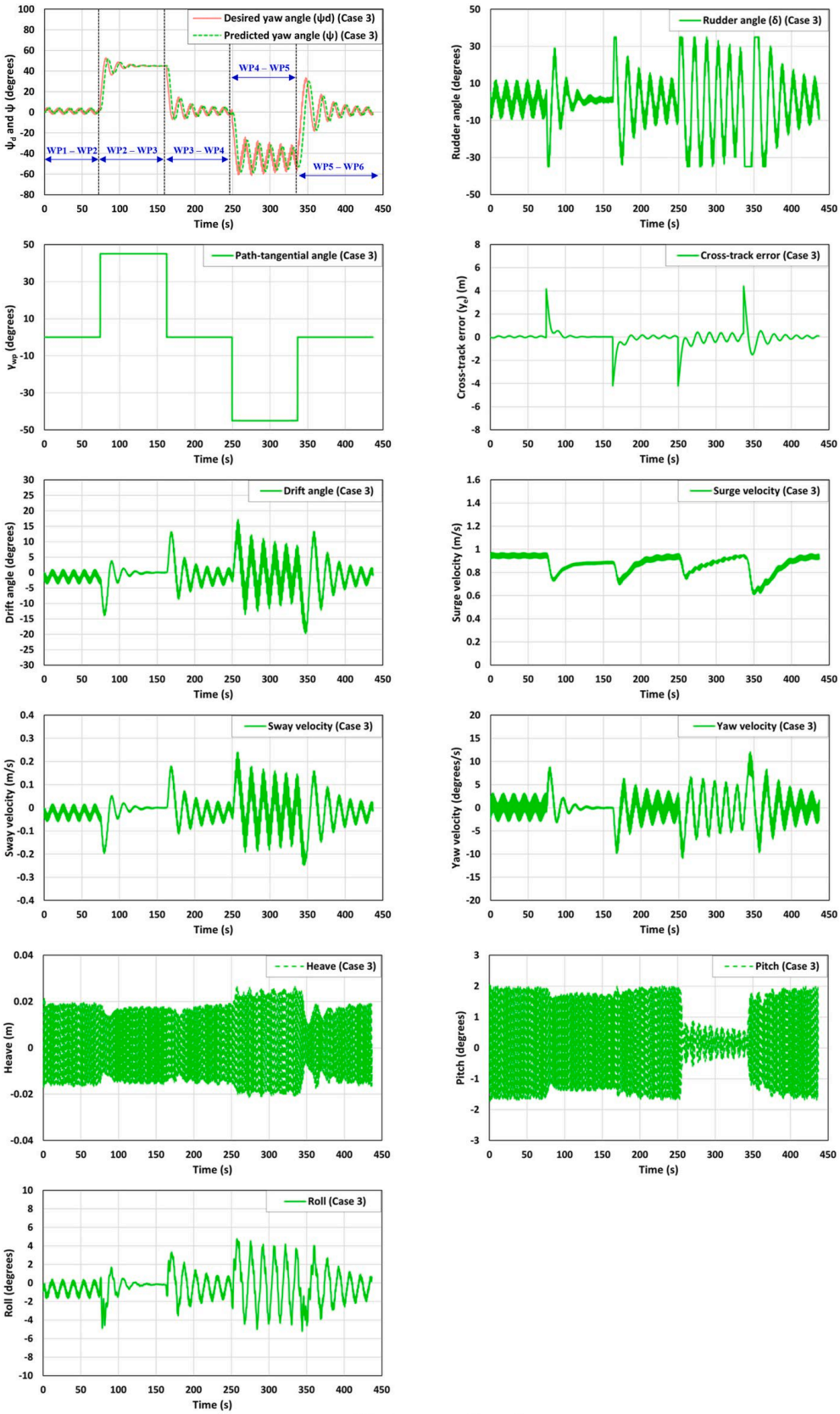


Fig. 15. *xy*-plot of the predefined and predicted ship paths during the path-following manoeuvre in waves of different directions (Case 2–6).



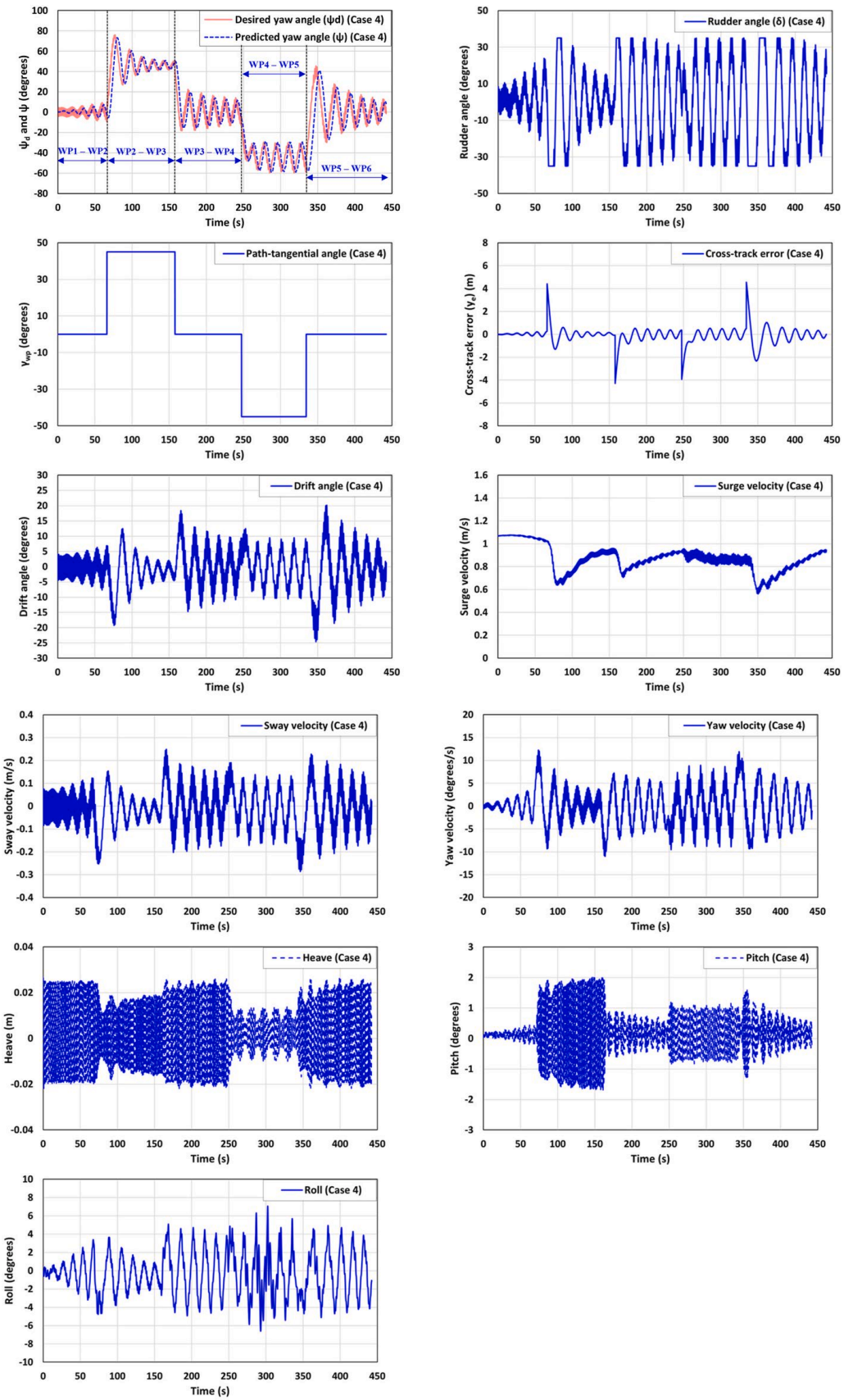
(a) Head sea (Case 2)

Fig. 16. The time histories of the critical LOS guidance parameters, kinematic parameters, and ship motions during the path-following manoeuvre in waves of different directions (Cases 2–6).



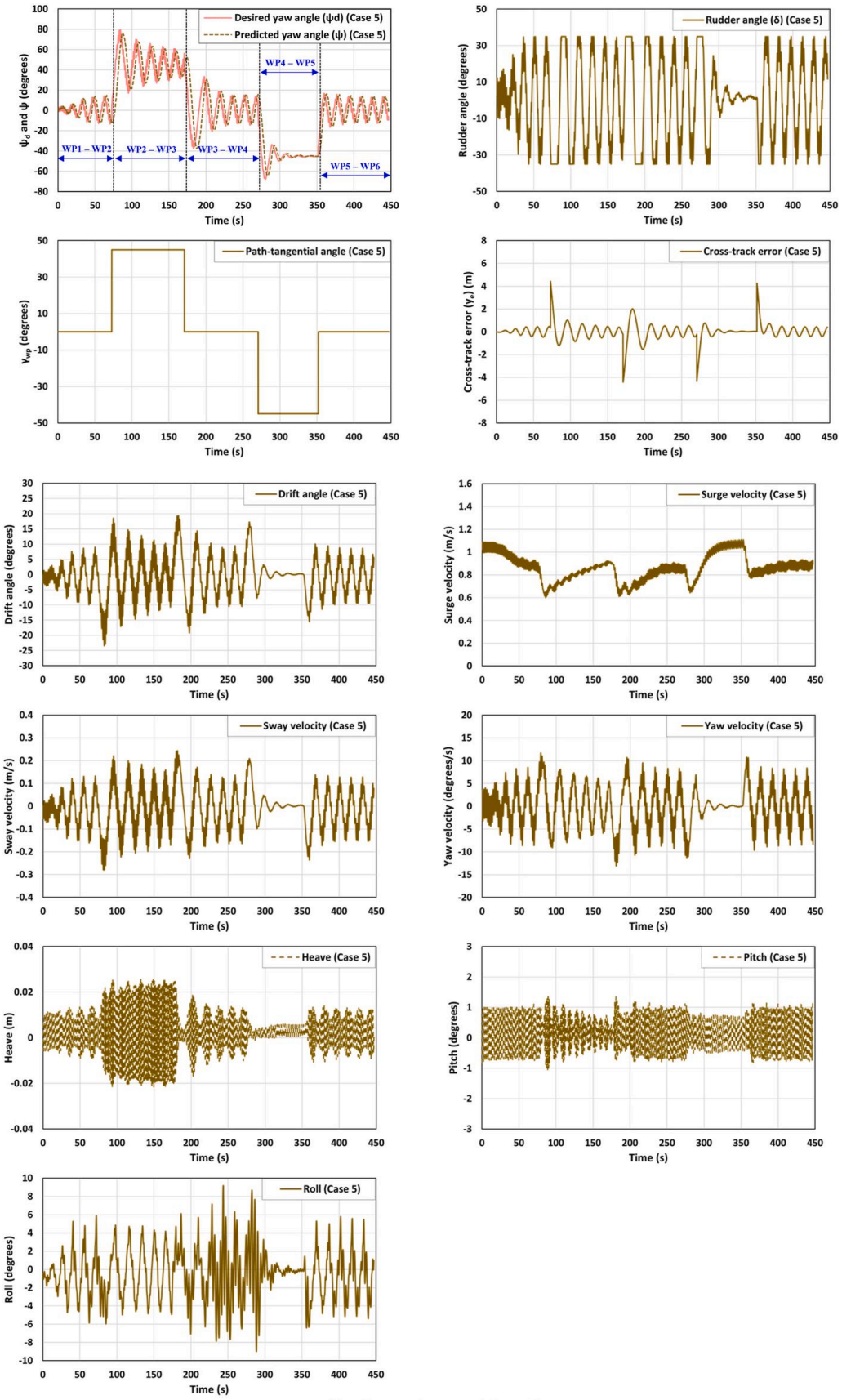
(b) Bow sea (Case 3)

Fig. 16. (*continued*).



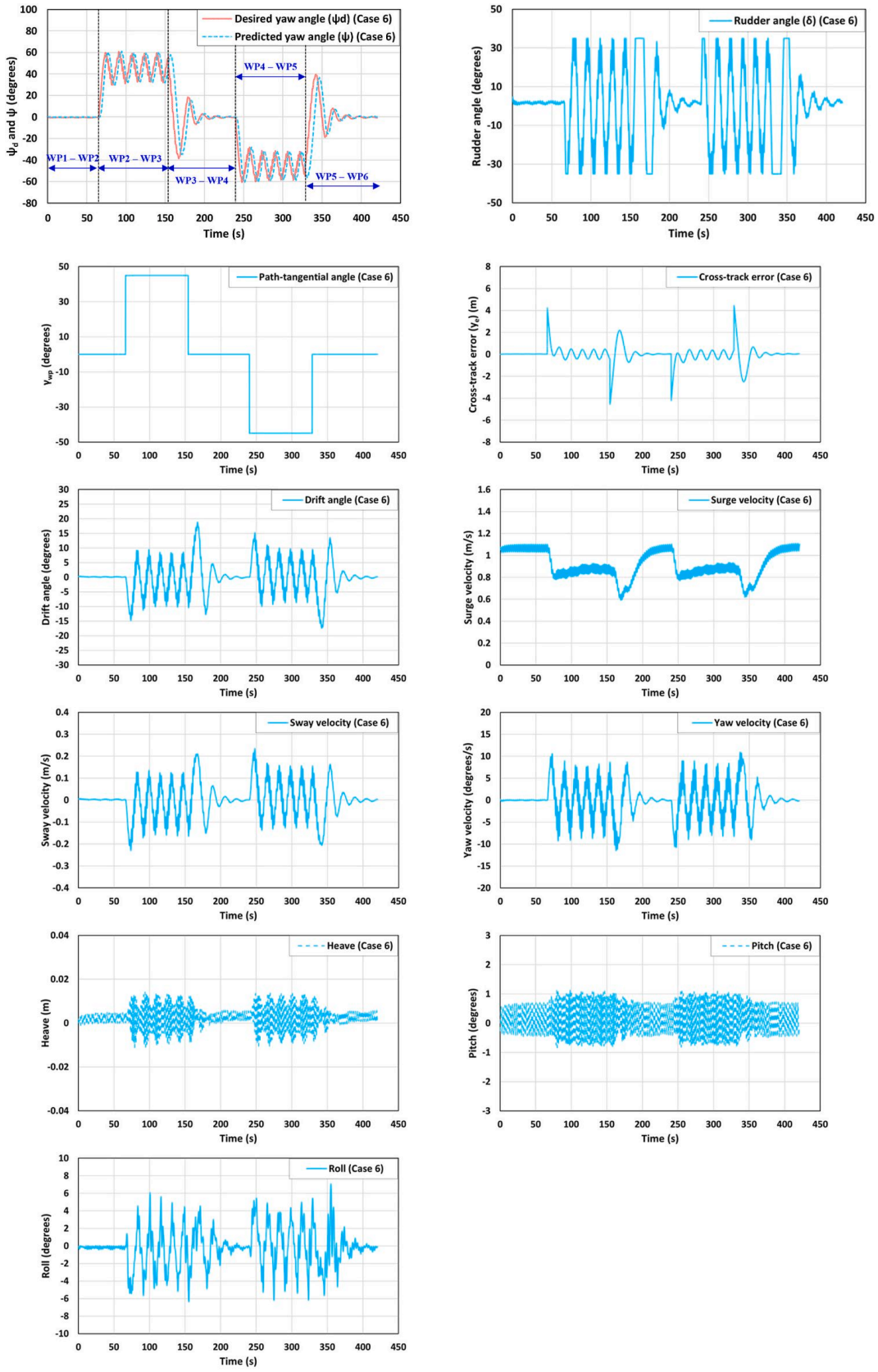
(c) Beam sea (Case 4)

Fig. 16. (*continued*).



(d) Quartering sea (Case 5)

Fig. 16. (*continued*).



(e) Following sea (Case 6)

Fig. 16. (*continued*).

should be noted that in all the CFD simulations, the propeller rotational speed was kept constant during the path-following task. In Case 2 (WP1-WP3), for example, the initial approach speed was observed to be 0.86 m/s, but after the course change at WP2, it sharply dropped to 0.69 m/s and then slowly recovered to 0.95 m/s which is greater than the initial speed. In contrast, recovered speeds after drastic speed losses were estimated to be greater than initial approach speeds in Cases 3, 4, and 6 (WP1-WP3). Such different trends in speed recoveries after the course change can be explained by the differences in the added resistance depending on the encounter-wave direction. In addition to this, it has to be noticed that the sway and yaw velocities experienced by the ship were mainly correlated to the pressure distribution on the ship caused by incident waves and the rudder normal force generated when the rudder was deflected. Relatively large fluctuations in sway and yaw velocities occurred as expected when the ship followed the predefined straight-line under the oblique and beam seas due to the asymmetric pressure distribution on the hull and oscillatory rudder behaviour with a large deflection angle.

Perceptible heave and pitch motions were observed in Cases 2–6 due to the presence of incident waves, as opposed to the calm water case (Case 1). The wave-encounter direction was recognised as the primary factor influencing notable changes in such motions during the path-following manoeuvre (for constant wavelength and wave height). According to the graphs illustrating the heave motion, the incident beam waves caused the maximum vertical motion. It can be explained by the ratio of wavelength to ship length; when the ship encountered the beam waves, its breadth is a relevant length quite smaller than the wavelength. Such a small relevant length may induce the maximum heave motion that is almost identical to the height of the incident wave. It is also worth pointing out that the heave and pitch responses of the ship to incident waves become larger due to the encounter frequency (f_e) being equal or close to the natural frequency of the heaving motion (f_n). According to a study by Kim et al. (2021c), the heaving and pitching system's natural frequencies for the KCS model are close to $f_n \approx 0.93$ Hz, in which it is expected that a larger excitation force will occur. Accordingly, when the ship encountered the head and bow seas ($f_e = 0.99$ and 0.93 Hz), the heave motions were observed to be relatively larger than those in the quartering and following seas ($f_e = 0.47$ and 0.37 Hz) due to the small difference between the encounter frequency and the natural heave frequency. In the same manner, the ship encountering the head and bow seas during the path-following control experienced relatively larger pitch responses when compared to those in the quartering and following seas. It can also be seen that the incident beam waves resulted in the minimum pitch motion by the fact that the waves coming from the beam did not induce the difference in pressure between the ship's bow and stern, consequently not generating the pitch moment. It was found that the roll motion experienced by the ship was closely related to the pressure distribution on the ship by incident waves and the rudder lift induced by the rudder deflection. When the ship followed the predefined straight-line in the beam and quartering seas, relatively larger roll motions were observed due to the asymmetric pressure distribution between the starboard and the port of the ship hull induced by the incident waves and the rudder normal force by large rudder deflections. In addition to this, large roll motions occurred when the ship changed its course towards the next waypoint throughout all the cases (Cases 2–6), resulting from the large rudder lift generated by the deflected rudder.

3.4. Path-following manoeuvres in waves of different heights (cases 7–11)

In this sub-section, the CFD results for the path-following control of the ship operating in waves of a range of wave heights (Cases 7–11) for constant wavelength will be given in detail. As reported in Table 2, the initial wave-encounter direction (when the ship started to perform the path-following manoeuvre) was the bow wave direction ($\mu = 225°$). Fig. 17 illustrates the *xy*-plot of the reference path and predicted trajectories experienced by the ship executing the path-following task in waves of different heights, together with zoomed-in views of the ship paths near the waypoints for clarity. The figure highlights the fact that the ship operating based on the LOS guidance algorithm successfully followed the predefined path under different wave height conditions from a practical point of view. It is also confirmed from the figure that wave heights strongly affected the ship trajectories under path-following control. When the ship carried out the path-following control with true course 000° (e.g., WP1-WP2, WP3-WP4, and WP5-WP6), deviation from the predefined path was found to increase with the increase in the wave height. The maximum deviation from the predefined path at the stabilised phase was estimated to be about 1.2m (corresponding to 0.4 L_{BP}) in the largest wave (H = 0.096m). In addition, the direction to which the ship mainly drifted was observed to be similar to the wave propagation direction. As was evidenced in the previous sub-section, path deviations

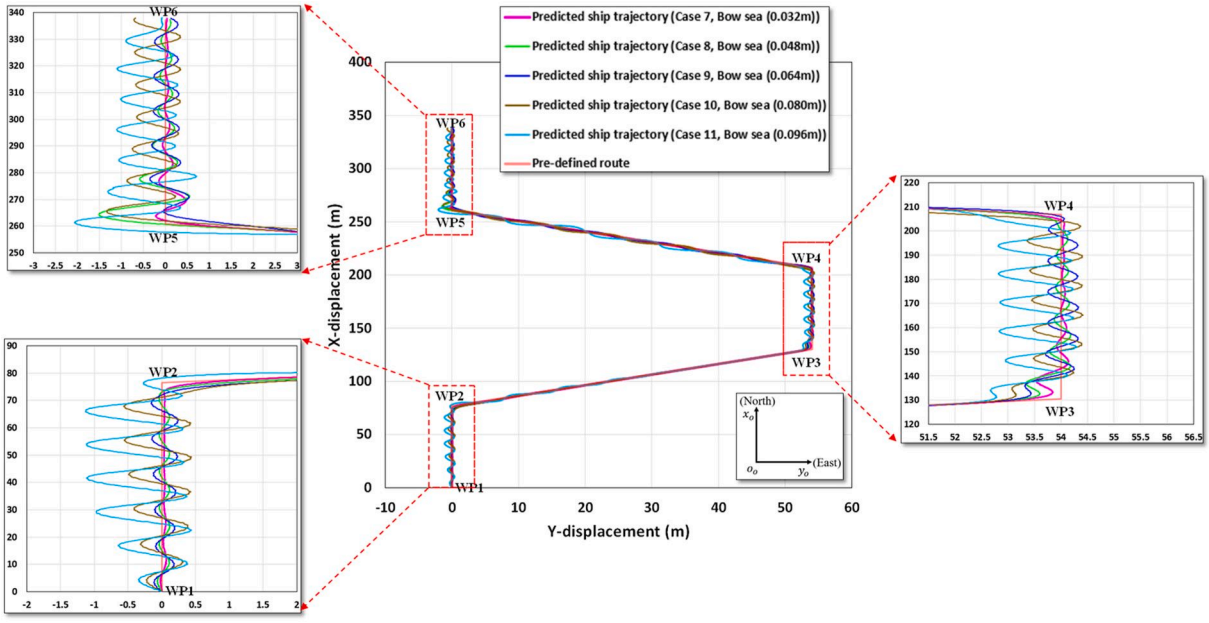
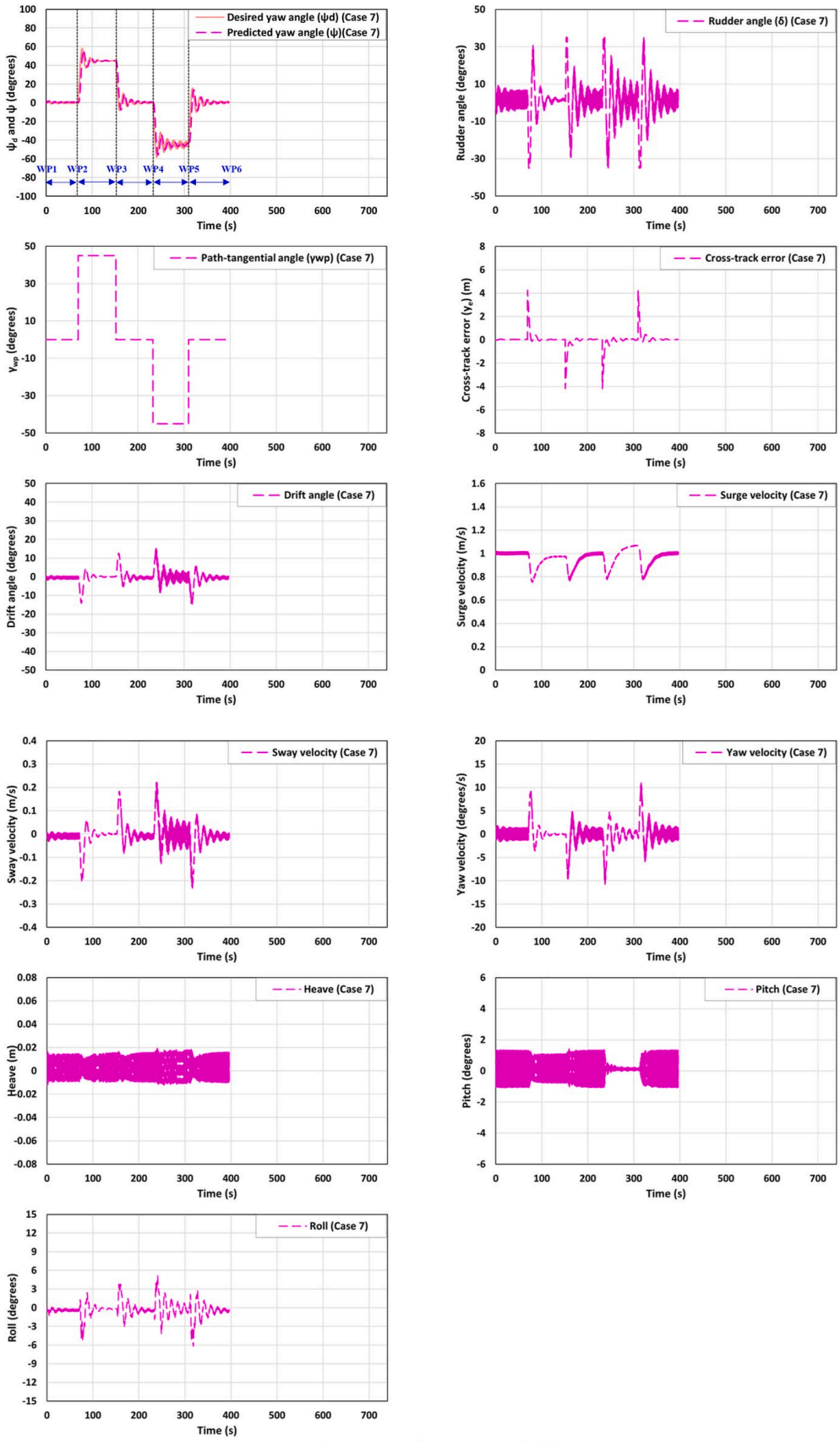
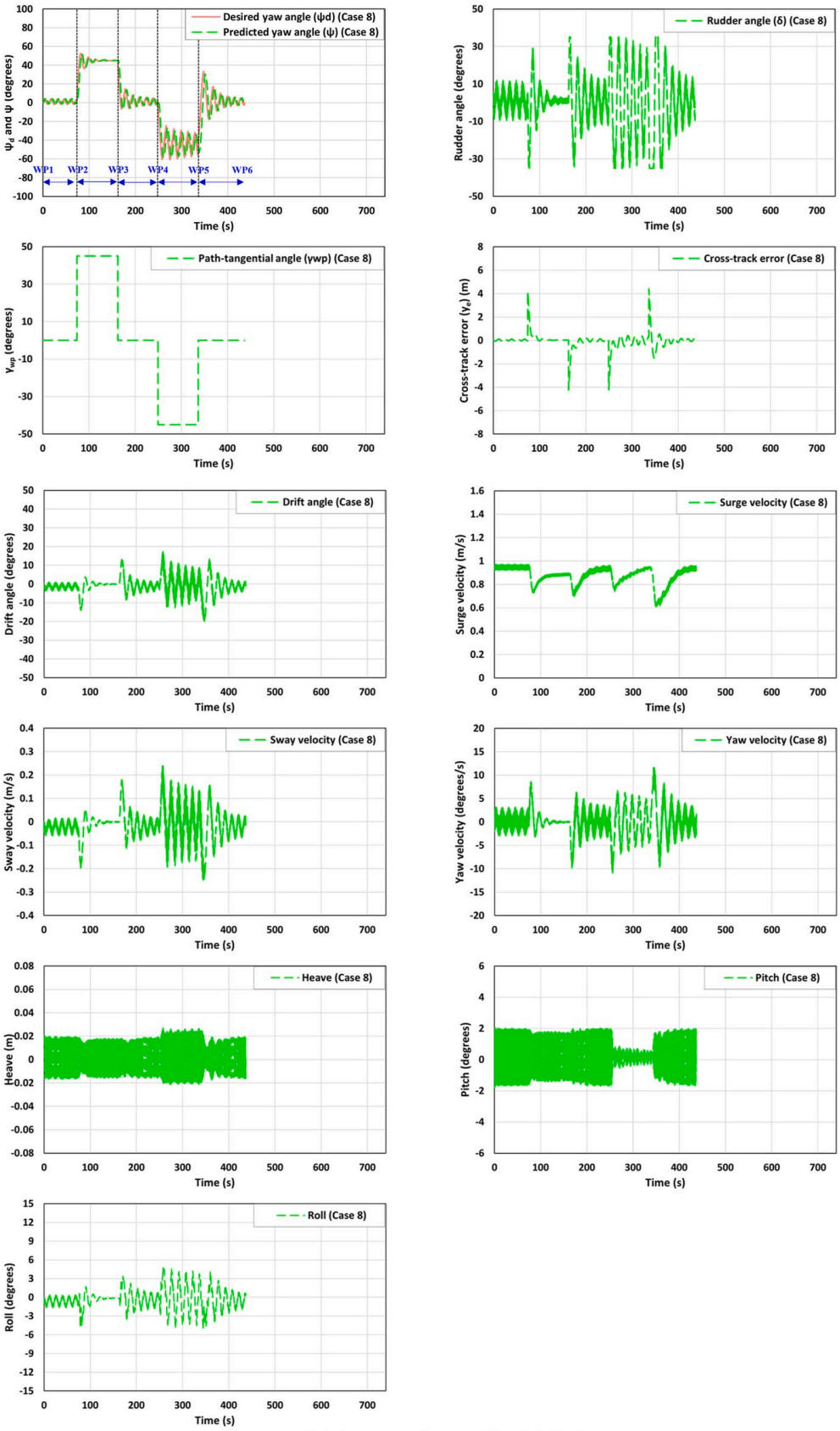


Fig. 17. xy-plot of the predefined and predicted ship paths during the path-following manoeuvre in waves of different heights (Cases 7–11).



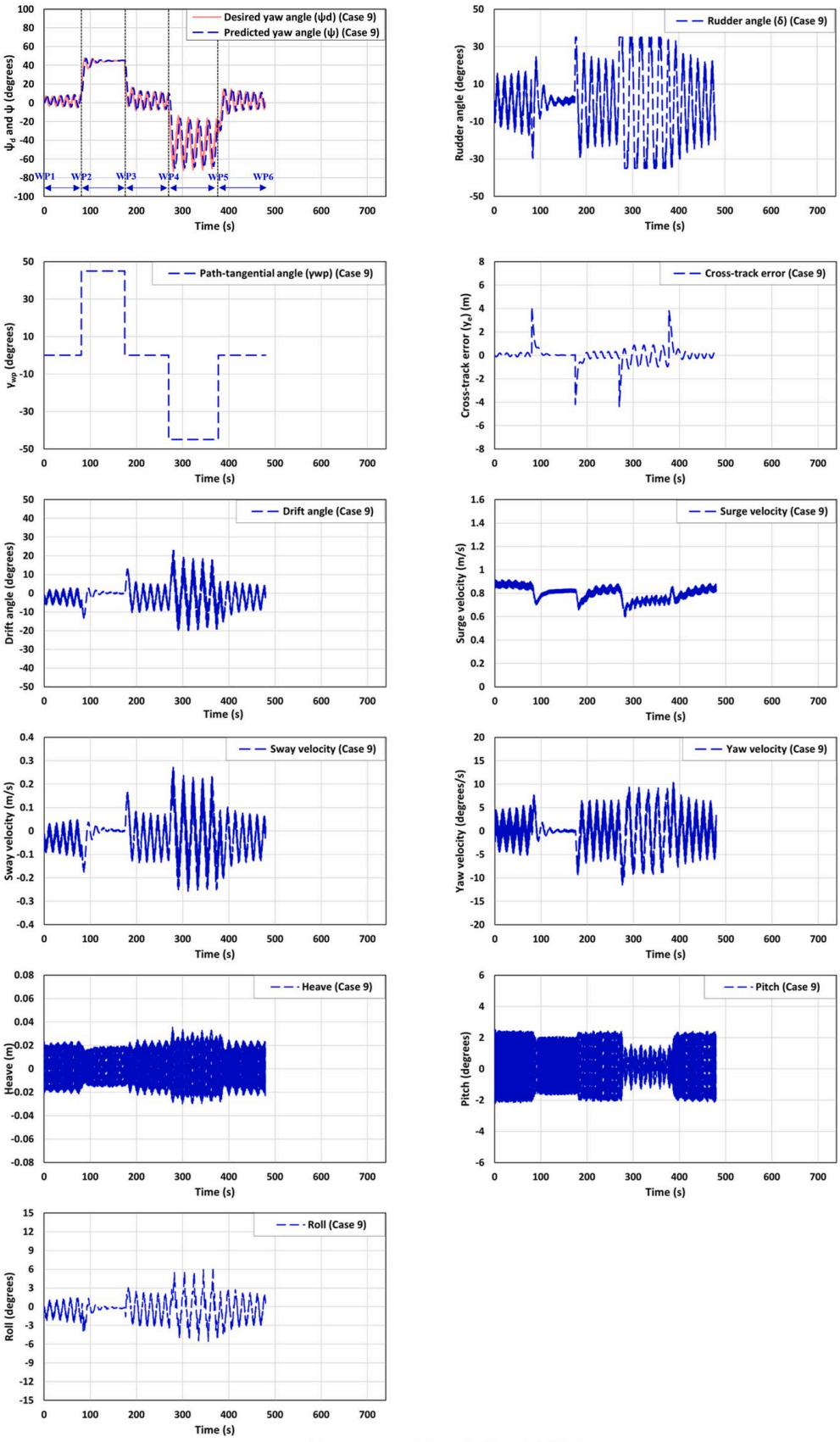
(a) Bow sea (Case 7, H = 0.032m)

Fig. 18. The time histories of the critical LOS guidance parameters, kinematic parameters, and ship motions during the path-following manoeuvre in waves of different heights (Cases 7–11).



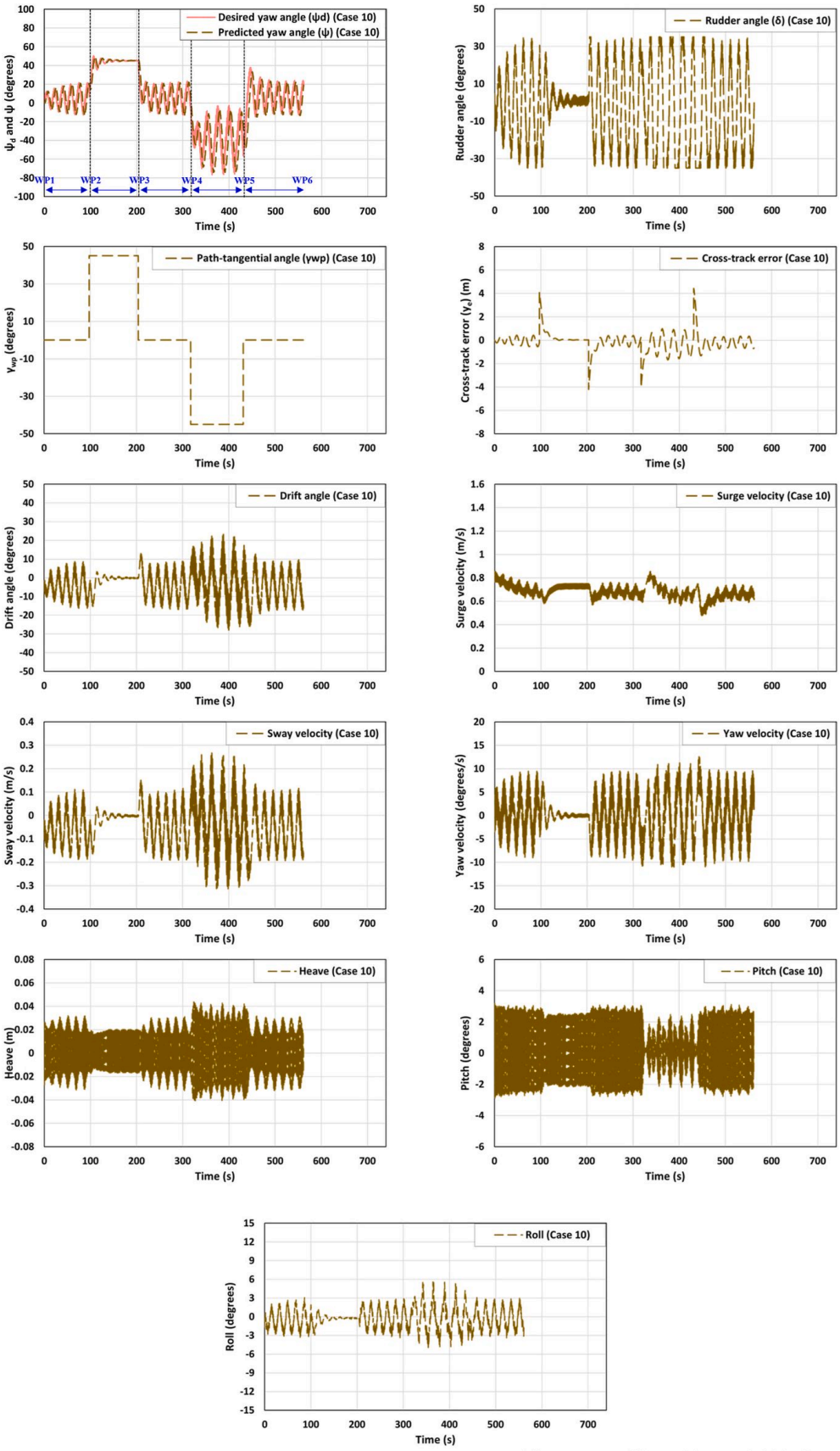
(b) Bow sea (Case 8, H = 0.048m)

Fig. 18. (*continued*).



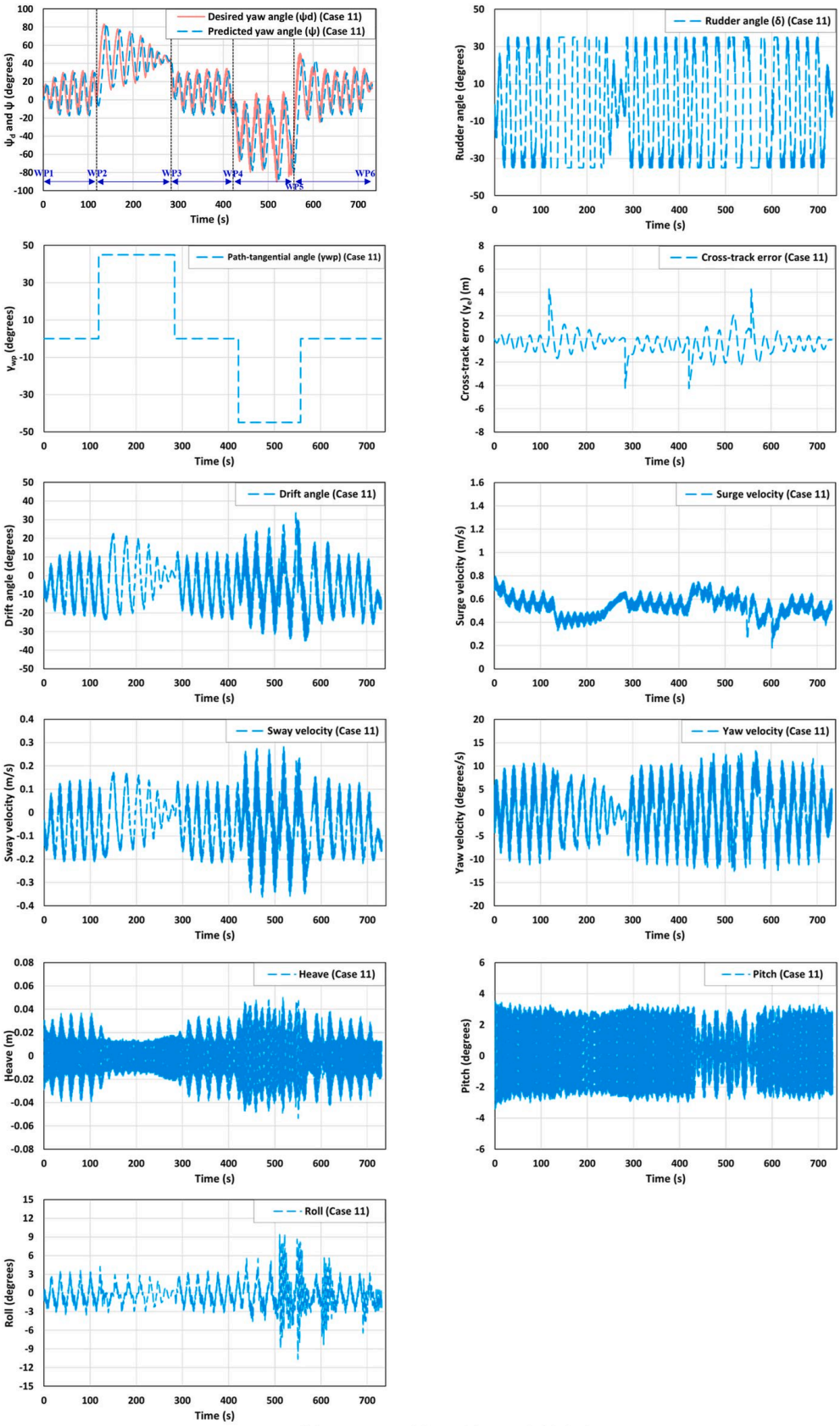
(c) Bow sea (Case 9, H = 0.064m)

Fig. 18. (*continued*).



(d) Bow sea (Case 10, H = 0.080m)

Fig. 18. (*continued*).



(e) Bow sea (Case 11, H = 0.096m)

Fig. 18. (*continued*).

in all wave height conditions were almost negligible in the head seas (WP2-WP3) because of the symmetric pressure distribution around the ship hull. On the contrary, when the wave-encounter direction was the beam seas (WP4-WP5), relatively larger path deviations were noticed due to the asymmetric pressure distribution on the ship hull.

Fig. 18 presents the time histories of the critical parameters pertaining to the LOS guidance law, kinematic parameters, and ship motions estimated during the path-following task in waves of different heights. When the ship performed the path-following control in the bow and quartering seas (WP1-WP2, WP3-WP4, WP4-WP5, and WP5-WP6), it was found that the oscillatory amplitudes of the desired yaw angle increased with an increase in the wave height and thus the amplitude of the predicted yaw angle appeared to be larger in the higher wave height. It was also evidenced from the graphs for the rudder angle that such oscillatory trends of the yaw angle are decisive for the behaviour of the rudder deflection. For the cross-track error, larger wave heights resulted in larger oscillatory amplitudes of the cross-track error in the oblique seas, which are contributors to the larger oscillatory desired yaw angle in the higher waves. It was seen from the graphs for the drift angle that the maximum drift angle occurred in the highest quartering waves, being consistent with the results reported in the previous sub-section.

When the ship followed the straight-line at the stabilised phase, the surge velocities reached the maximum when the ship experienced the quartering seas (WP4-WP5), whereas the minimum values were predicted under the head seas (WP2-WP3) due to the difference in the added resistance throughout all the cases (Cases 7–11). It was also clearly seen that the increase of the wave height led to a decrease in the surge speed when the ship performed the path-following manoeuvre with the same true course. This is mainly due to the fact that the ship resistance has a proportional relation with the added wave resistance which is generally proportional to the wave height squared. A larger wave height induced a larger pressure difference between the starboard and the port of the hull (i.e., asymmetric pressure distribution) during the path-following task under the oblique seas (WP1-WP2, WP3-WP4, WP4-WP5, and WP5-WP6). As a result of this, larger rudder deflections were observed in the larger wave height conditions, consequently leading to larger oscillatory sway and yaw velocities.

It can be revealed that when the ship followed the same reference straight-line (with the same true course), the heave and pitch excitation force and moments increased with an increase in the wave height. Thus, the heave and pitch responses of the ship appeared to become larger under the larger wave height conditions, as clearly evidenced in the graphs for the heave and pitch motions. For the roll motions, it was also demonstrated that larger wave heights can result in comparatively larger roll amplitudes during the path-following task, mainly due to the aggressive rudder deflection leading to the large rudder lift.

4. Conclusions and discussion

In the present work, a CFD-based dynamic model for path-following problems has been developed for autonomous marine navigation by means of a fully nonlinear unsteady RANS solver combined with the LOS guidance law. Given that an accurate prediction of the path-following performance for remote-controlled ships or maritime autonomous surface ships (MASS) is of critical importance to ensure safe autonomous navigation, it is expected that this paper will add a sound value to the field of autonomous vehicles by presenting the effectiveness of the proposed CFD model. To obtain practical insight into the performance of the ship executing path-following control for autonomous navigation in waves, a comprehensive analysis of path-following tasks in waves covering the whole range of important wave directions and wave heights has been carried out. The key findings of this study can be summarised as follows.

(1) It was identified that the application of Wheel Over Point to the present CFD model performing the path-following task based on the LOS guidance law can decrease the deviation from the predefined path while changing its course towards the next waypoint.
(2) It was revealed that the path-following task in calm water (Case 1) was not an issue due to the absence of external disturbances, presenting good performance of the straight-line path following control as well as the course alteration to the starboard/port side.
(3) The analysis of the path-following task under different wave direction conditions (Cases 2–6) demonstrated that wave directions provided a noticeable contribution to the ship trajectories under path-following control. When the ship performed the path-following manoeuvre under the beam and quartering seas, the greatest oscillatory deviation from the predefined path was observed due to the asymmetric pressure distribution on the ship hull caused by the incident waves.
(4) It was confirmed that wave heights strongly affected the trajectories experienced by the ship performing the path-following task in the beam and oblique seas; the deviation from the predefined path was found to increase with an increase in the wave height. On the contrary, path deviations in all wave height conditions were almost negligible in the head seas due to the symmetric pressure distribution around the ship hull.

Deviations from the predefined path in voyage plans should be recognised as a critical safety factor when establishing voyage plans in terms of the path-following task for autonomous marine navigation. As comprehensively discussed in this paper, it is expected that autonomous maritime vehicles exposed to waves experience larger deviations from the predefined path when compared to those in calm water due to the external forces and moments induced by incident waves. Such deviations can potentially lead to navigational casualties such as grounding and collision posing serious threats to safe autonomous at sea. It is therefore critical to accurately predict the performance of path-following tasks in the area where an autonomous vehicle is to be navigated through. A safe navigation route in the voyage plan should be determined in careful consideration of the path-following performance predicted before a voyage commences. From this point of view, it is highly believed that the CFD-based dynamic model proposed in this paper can provide a valuable contribution to enhancing the safety of autonomous marine navigation. In addition, it is expected that the proposed CFD model will allow a control system for path-following tasks to be evaluated with an acceptable level of reliability, offering low-cost alternatives to model-scale free-running experiments or full-scale sea trials.

It should be noted that the CFD approach used in this study is significantly more time-consuming than conventional system-based free-running simulations. In addition, this approach will necessitate new CFD simulations whenever a different type of path-following manoeuvre needs to be executed. As high-performance computational resources become increasingly available, however, the proposed CFD method will provide an accurate and efficient way to predict the performance of autonomous surface ships performing path-following tasks in waves.

The research tackled in this paper has provided a practical starting point for the examination of a ship's path-following performance in calm water and regular waves by means of an unsteady RANS method. However, it has to be emphasised that irregular waves can better represent the real operational conditions at sea when compared to both calm and regular seas. Given this, this study could further add value by investigating the contribution of irregular waves to the path-following performance for autonomous marine navigation in a real seaway by developing a free-running CFD model for path-following control in irregular waves.

CRediT authorship contribution statement

Daejeong Kim: Writing – original draft, Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Data curation. **Soonseok Song:** Methodology, Writing – review & editing, Resources. **Stephen Turnock:** Writing – review & editing, Resources. **Tahsin Tezdogan:** Supervision, Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was supported by the 'Development of Autonomous Ship Technology (20200615)' funded by the Ministry of Oceans and Fisheries (MOF, Korea). It should be noted that the results were obtained using the ARCHIE-WeSt High Performance Computer (www.archie-west.ac.uk) based at the University of Strathclyde.

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